Retention-constrained class expansion with reference-anchored candidate tables

Junyi Zhang[a,*]

[a] *Shandong University, China*

Abstract

Expandable recognition must admit new categories while preserving an explicit accuracy requirement on the original classes. We study novel recognition at a common original-class constraint and the predictive value of retaining a classifier across successive arrivals. Reference-anchored candidate tables combine trained base columns, semantic anchors and novel visual prototypes in a persistent score frame. Base-only pseudo-expansion episodes calibrate the mixtures; validation-selected admission thresholds are evaluated separately from descriptive matched-test frontiers. Experiments cover CIFAR-100, Oxford-IIIT Pets and 200-class ImageNet-R, ten class partitions per dataset and two frozen CLIP architectures. On ImageNet-R with ViT-L/14, a common base accuracy of 86.42% permits 84.31% novel accuracy. Across five arrivals, retaining the classifier adds 0.71 percentage points over dense prototype fusion and 1.42 points over class-mean semantic-cache retrieval in average incremental accuracy. The prototype comparison transfers to ViT-B/32 with a 0.90-point gain. Paired confidence intervals characterize class-partition variation conditional on the fixed data and encoders. Raw-score and fixed-temperature controls isolate reference normalization, while complete validation-policy results distinguish empirical feasibility from attained test retention. The evidence identifies the operating regimes in which preserving trained discrimination supplies additional class-expansion capacity.

Keywords: Class expansion, Retention constraints, Incremental learning, Evidence fusion, Vision–language models, Long-tailed recognition

1. Introduction

Recognition systems often acquire new categories after deployment, while the original classifier has been fitted to imbalanced supervision. Long-tailed recognition addresses the

*Corresponding author.
Email address: cheerssom0876@gmail.com (Junyi Zhang)

unequal evidence available to common and rare classes [1, 2, 3]. Class expansion must put that learned discrimination to work alongside the names and sparse examples supplied for arriving categories.

Pre-trained vision–language models provide a shared interface for this evidence. CLIP maps images and descriptions into an embedding space [4]; prompt adaptation and lightweight fine-tuning specialize its predictions [5, 6, 7]; caches and prototypes incorporate labeled support [8, 9, 10]. An expanded classifier can therefore retain trained base columns and append semantic anchors and prototypes for novel classes.

The practical criterion is novel recognition under a stated base-accuracy requirement. Fixing this requirement identifies how much new-class recognition the system can admit while preserving its established service level. Successive class arrivals introduce a second criterion: the paired gain obtained by retaining the base-trained classifier throughout the sequence. Separating a fixed-gate removal from a recalibrated classifier-free alternative measures both the contribution of the stored learned evidence and its value after calibration is restored.

We address these criteria with *reference-anchored candidate tables*. Each class stores the evidence available to it, and each branch normalizes its scores against fixed base-class references. The table retains the original learned columns and appends new rows without refitting them. This reference frame makes the existing scores independent of vocabulary size, allowing the novel admission threshold to be analyzed separately from within-group discrimination.

Class-incremental learning studies retention of earlier knowledge [11, 12], and pre-trained prototypes and analytic readouts enable efficient class addition [13, 14]. Recent methods combine multiple feature spaces [15], calibrate prototypes through sharpness-aware learning [16], and fuse CLIP prompts across modalities [17]. Our evaluation asks what a retained classifier contributes under a common base constraint and across repeated arrivals. A score-preservation proposition supplies the mechanism; matched-retention frontiers and paired sequential comparisons supply the recognition evidence.

On the 200-class ImageNet-R task, the retained classifier adds 0.71 [0.46, 0.92] percentage points in average incremental accuracy over dense prototype fusion with ViT-L/14 and 0.90 [0.69, 1.11] with ViT-B/32. Brackets are paired 95% bootstrap intervals across ten class partitions, conditional on the fixed image pools and encoders. The expanded comparison separates classifier removal, recalibration, prototype replacement, semantic-cache retrieval and simpler score fusion. Validation-selected thresholds and their achieved test outcomes

receive a separate main-paper comparison.

The contribution is a retention-constrained fusion and evaluation framework: it organizes base-discriminative and arriving evidence in a persistent score frame, separates within-base discrimination from the admission threshold, and measures the retained classifier's paired value against recalibration and replacement. Raw and fixed-temperature controls isolate the role of the per-query reference statistics; expanded class-partition and encoder comparisons identify the operating regimes in which retaining trained evidence is useful.

2. Related work

2.1. Imbalanced supervision and heterogeneous evidence

Long-tailed recognition motivates balanced fitting when class evidence is unequal [2, 3]. CLIP supplies semantic anchors [4], prompt learning adapts textual evidence [5, 6], and prototypes and caches incorporate labeled support [9, 10, 8]. Our table combines these evidence types according to their availability, with a shared encoder and novel support budget across compared readouts.

2.2. Few-shot expansion and retained discrimination

Multi-feature space similarity supplement (MFS3) combines base and incremental feature spaces to refine boundary decisions [15]. Sharpness-aware gradient guidance (SAGG) couples flat-region optimization with prototype calibration [16]. Adaptive exploration uses reward-guided updates to balance knowledge retention and acquisition [18]. These approaches develop representations or classifiers for successive tasks. Our formulation evaluates the recognition capacity supplied by an existing classifier at a shared base-accuracy requirement. Fixed-gate removal, independent recalibration and dense prototype replacement identify distinct sources of its measured value.

2.3. Incremental recognition with pre-trained models

Class-incremental learning evaluates a model on the union of classes observed across successive training stages. Exemplar rehearsal and representation constraints are established tools for retaining earlier knowledge [11]. Few-shot class-incremental learning adds a data scarcity condition to the arriving classes [12]. Recent work studies temporal consistency with unlabeled memory replay [19] and the geometry of inter-task and intra-task discrimination [20]. These approaches motivate explicit measurement of existing-class behavior.

SimpleCIL examines prototype classifiers built from pre-trained representations [13]; RanPAC combines random projections with accumulated class statistics [14]. Reference-anchored fusion extends this setting to semantic, trained, and support-derived evidence with different availability patterns. Calibrated seen/unseen recognition [21] provides a related perspective on the base–novel operating trade-off.

Bimodal Prompt Collaborative CLIP combines attribute-enriched text with gated visual–textual prompts for few-shot incremental learning [17]. CLIP-guided continual novel class discovery uses semi-supervised prototype adaptation to transfer visual–language knowledge to a discovery model [22]. The latter operates with unlabeled novel data; our arriving classes have names and labeled support. REAL enhances representations for exemplar-free analytic learning through complementary base pretraining and feature fusion [23]. These studies establish complementary ways to retain and combine prior knowledge. Reference-anchored tables organize semantic anchors, learned columns and support prototypes within one fixed score frame, making the retained classifier's admission value directly measurable.

FeNeC represents classes with clustered feature representatives and class-specific distances, extending nearest-class-mean recognition to local neighborhoods [24]. Its Mahalanobis and Euclidean variants provide recent frozen-feature comparisons under the same base fitting data and five-shot novel support. We report native predictions and an explicitly group-offset-calibrated form.

BiMC [25] calibrates textual classifiers with fine-grained descriptions, adjusts novel visual prototypes using base prototypes, and combines modality and covariance-based metrics. HyCal [26] combines cosine and Mahalanobis information to address heterogeneous cross-discipline few-shot streams. These methods improve the evidence or metric used for incremental recognition. Our framework instead measures the contribution of a separately fitted base-discriminative branch under a shared retention requirement and preserves an explicit score frame while heterogeneous evidence is appended. The relevant comparison is the attainable base–novel trade-off after equalizing data access and calibration, rather than an unrestricted accuracy comparison across different protocols.

TV100 studies categories unavailable during CLIP's original pre-training period [27]. Here, novelty is defined relative to task-specific classifier fitting and calibration; CLIP retains its web-scale pre-training exposure.

3. Expandable recognition with heterogeneous evidence

3.1. Problem definition

Let $\mathcal{B}$ denote base classes with a labeled training set $\mathcal{D}_B$, and let $\mathcal{N}$ denote novel classes supplied at expansion time. The active vocabulary is initially $\mathcal{C}_0 = \mathcal{B}$ and subsequently becomes $\mathcal{C} = \mathcal{B} \cup \mathcal{N}$. An image encoder f maps an input image to a normalized vector $v = f(x) \in \mathbb{R}^D$. A text encoder supplies a normalized class anchor t_c. For each $c \in \mathcal{N}$, the support set $\mathcal{S}_c$ contains k labeled examples. Inference is over all active classes, without an oracle indicating whether an image belongs to a base or novel class.

Base training can be imbalanced. If n_c is the number of training images for class c, the classifier may use inverse-frequency weights or another balanced fitting procedure. After this stage, the encoder, base classifier, reference banks, and branch mixing coefficients are fixed. The group admission offset is either frozen or calibrated on base validation to enforce a retention target. At arrival, novel class information is restricted to its name and $\mathcal{S}_c$. Neither previous training images nor additional labeled novel validation images are required to construct its row.

Each row has an availability mask $m_{ec} \in \{0, 1\}$ over evidence sources e. We use semantic evidence T, learned discriminative evidence L, and visual prototype evidence P. The experimental configuration is

$$(m_{Tc}, m_{Lc}, m_{Pc}) = \begin{cases} (1, 1, 0), & c \in \mathcal{B}, \\ (1, 0, 1), & c \in \mathcal{N}. \end{cases} \tag{1}$$

Base prototypes form the visual branch's normalization references. The availability mask determines which evidence contributes to a candidate; the reference bank determines its scale.

3.2. Three notions of preservation

Preserving a parameter vector θ means that θ receives no update during expansion. Preserving a score additionally requires

$$s_{\mathcal{C}_0}(x, c) = s_{\mathcal{C}}(x, c), \quad c \in \mathcal{C}_0. \tag{2}$$

Preserving a prediction requires the maximizing class over the larger vocabulary to remain the same. The third property is stronger than the second: a newly admitted class can have a larger score than the previous winner. We seek score preservation while allowing that competition.

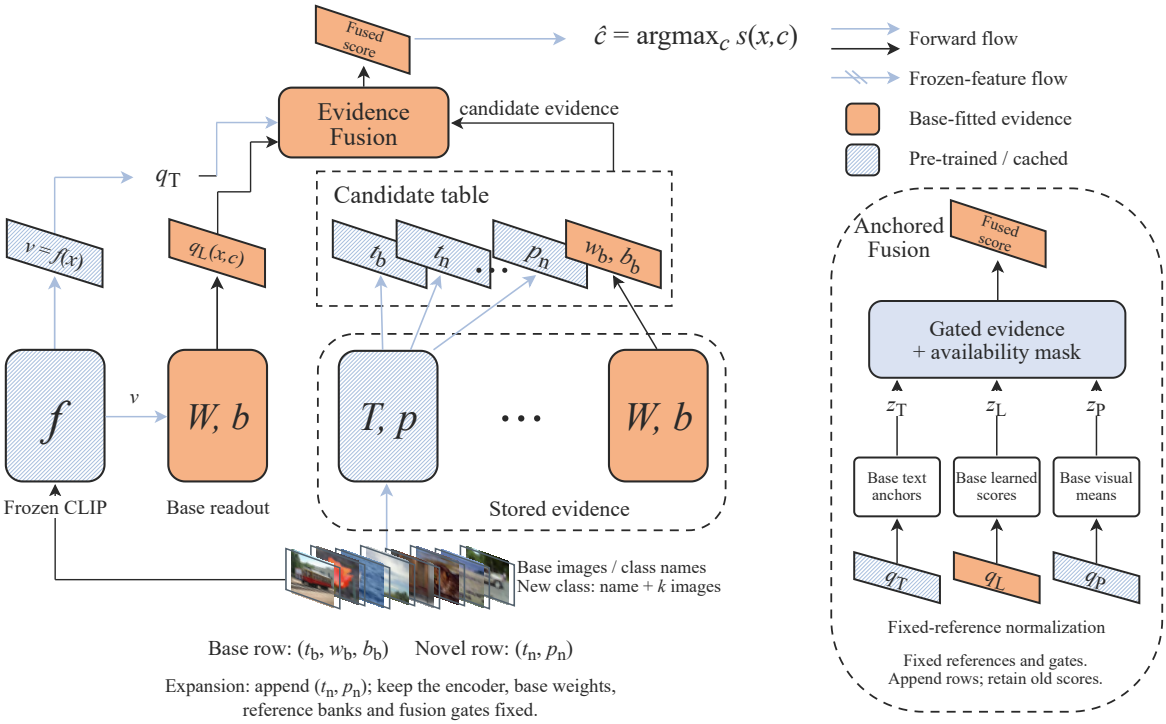


Figure 1: Reference-anchored candidate-table architecture. A shared frozen CLIP feature supplies semantic similarity and the base-trained readout. The candidate table stores text anchors for all classes, learned weights for base classes, and support prototypes for arriving classes. The inset shows the three evidence branches normalized against their respective fixed base reference sets before availability-masked fusion; $z_e = \widehat{q}_e$ denotes normalized evidence. A novel row is appended without updating the encoder, existing weights, references, or gates. Orange identifies base-fitted evidence and fusion; blue hatching identifies pre-trained or cached evidence. Branch gates remain fixed; retention control can adjust the separate novel admission offset.

The evaluation separates changes among existing classes from selection of an appended class. Under fixed evidence, reference anchoring preserves existing-class competition and admits new alternatives through their own scores.

4. Reference-anchored candidate tables

4.1. Evidence construction

The semantic score is cosine similarity between a normalized image feature and a class text anchor,

$$q_T(x, c) = v^\top t_c. \tag{3}$$

We average normalized text embeddings of eight fixed prompts and normalize the result. The same prompts construct anchors for base and arriving classes.

The learned score is the output of a base classifier,

$$q_L(x, c) = v^\top w_c + b_c, \quad c \in \mathcal{B}. \tag{4}$$

For the frozen-encoder experiments, the weights are fitted using class-balanced ridge regression. With augmented image features $\widetilde{v} = [v; 1]$, one-hot targets Y, and diagonal sample-weight matrix Ω, the solution is

$$W = (X^\top \Omega X + \lambda R)^{-1} X^\top \Omega Y, \tag{5}$$

where $R = \mathrm{diag}(1, \ldots, 1, 0)$ leaves the intercept unregularized. A sample from class c has weight $1/n_c$, giving each base class equal total fitting weight. We solve the corresponding linear system.

The visual prototype of a novel class is

$$p_c = \frac{\sum_{x \in \mathcal{S}_c} f(x)}{\| \sum_{x \in \mathcal{S}_c} f(x) \|_2}, \qquad q_P(x, c) = v^\top p_c. \tag{6}$$

The same averaging rule applied to base training features produces a reference bank $\{p_b : b \in \mathcal{B}\}$. The novel prototype stores the information acquired at class arrival; the base bank fixes the coordinate system in which its score is interpreted.

4.2. Why a changing normalization population matters

Candidate-wise standardization computes the mean and standard deviation over the current valid candidates of a branch. For text evidence, this gives

$$z_T^{\mathcal{C}}(x, c) = \frac{q_T(x, c) - \mu_T(x; \mathcal{C})}{\max(\sigma_T(x; \mathcal{C}), \epsilon)}. \tag{7}$$

Adding candidates changes μ_T and σ_T. For a text-only classifier, a common positive affine transformation preserves the ordering of the old classes. With heterogeneous fusion, however, the learned branch and text branch are rescaled differently. The relative strength of their votes can change.

A two-class example illustrates the effect. Suppose the old text scores are $(0, 2)$, giving standardized scores $(-1, 1)$, while the learned branch contributes $(0.25, -0.25)$. The second old class has the larger fused score. Appending a text score of 100 changes the two old standardized scores to approximately $(-0.728, -0.685)$. Their text separation is now about 0.043, and the unchanged learned contribution reverses the old-class ordering. No classifier weight has changed. The example isolates a normalization effect independently of whether the new class ultimately wins the expanded decision.

4.3. Fixed references for each evidence source

Let $\mathcal{R}_e$ be a fixed bank of reference rows for source e. For every query, compute

$$\mu_e(x) = \frac{1}{|\mathcal{R}_e|} \sum_{r \in \mathcal{R}_e} q_e(x, r), \tag{8}$$

$$\sigma_e^2(x) = \frac{1}{|\mathcal{R}_e|} \sum_{r \in \mathcal{R}_e} (q_e(x, r) - \mu_e(x))^2. \tag{9}$$

The normalized candidate score is

$$\widehat{q}_e(x, c) = \frac{q_e(x, c) - \mu_e(x)}{\max(\sigma_e(x), \epsilon)}. \tag{10}$$

The reference statistics remain query-dependent, which adapts the scale to the current image, but are independent of which novel candidates are active. We use base text anchors for $\mathcal{R}_T$, base classifier columns for $\mathcal{R}_L$, and base visual means for $\mathcal{R}_P$. Each bank contains 40 reference classes on CIFAR-100 and 15 on Pets. A zero-variance reference bank uses the same fixed denominator floor $\epsilon = 10^{-6}$.

The candidate score combines available normalized evidence:

$$s(x, c) = \sum_{e \in \{T, L, P\}} m_{ec} \alpha_e \widehat{q}_e(x, c) + \delta_c. \tag{11}$$

We set $\alpha_T = 1$, write $a = \alpha_L$ and $u = \alpha_P$, and use $\delta_c = d\,\mathbf{1}[c \in \mathcal{N}]$. The resulting implementation is

$$s(x, b) = \widehat{q}_T(x, b) + a\widehat{q}_L(x, b), \quad b \in \mathcal{B}, \tag{12}$$

$$s(x, n) = \widehat{q}_T(x, n) + u\widehat{q}_P(x, n) + d, \quad n \in \mathcal{N}. \tag{13}$$

The offset d calibrates competition between base and novel classes. The availability mask excludes absent learned evidence from the novel score and from reference statistics.

4.4. Score preservation under append-only expansion

Proposition 1. *Fix an input x, the evidence of every existing row, the encoder, the reference banks, the denominator floor, and the coefficients in Eq.* (11)*. Appending new rows leaves the score of every existing row unchanged. Hence, every pairwise score difference and the maximizing row restricted to the existing vocabulary are unchanged, using a fixed tie-breaking rule.*

Each existing raw score, availability mask and base-reference statistic stays fixed, so their weighted sum stays fixed. The proposition specifies this structural invariant; it does not by itself imply a recognition gain, and raw or fixed-temperature fusion can satisfy the same invariant. This applies to base and previously appended novel rows. Supplementary Section E provides the full proof and the consequence for expanded predictions. The implementation computes the fixed base text block separately from novel columns to keep its numerical kernel independent of the appended width.

4.5. Base-only calibration

Fusion calibration must respect the distinction between base supervision and novel support. We construct a pseudo-expansion episode by splitting the base classes into approximately equal pseudo-base and pseudo-novel groups: 20/20 on CIFAR-100, 7/8 on Pets and 50/50 on ImageNet-R. A separate classifier is fitted using the pseudo-base classes alone. The pseudo-novel rows contain names and k support examples from those base classes, while held-out base validation images supply query labels. This episode simulates missing learned evidence without consuming any actual novel-class validation labels.

For each support budget, candidate coefficients are evaluated on this episode. The selection objective is the harmonic mean of macro accuracy on the two groups,

$$H = \frac{2A_B A_N}{A_B + A_N}, \tag{14}$$

with $H = 0$ if both terms vanish. The main experiment uses $a \in \{0, 0.25, 0.5, 1, 2\}$, $u \in \{0, 0.25, 0.5, 1, 2, 4\}$, and offsets $d \in [-4, 4]$ in steps of 0.25. Ties follow a fixed grid order. The full base classifier is then combined with actual novel rows using the selected coefficients. Neither actual novel validation queries nor test labels enter this selection rule.

This procedure uses one pseudo-class split per true partition and the same support-index seed as the evaluated support condition. CIFAR and Pets pseudo splits use seed $100 + p$ for class-partition seed p; ImageNet-R selects half the base classes with seed $900 + p$. Coefficients

are chosen once for a support budget and stay fixed when the vocabulary-size stress test appends successive subsets of novel classes.

4.6. Storage and computation

The frozen implementation uses one image feature per input. Class text anchors, base classifier weights, visual references, and novel prototypes are cached. For feature dimension D, the base reference banks require $O(|\mathcal{B}|D)$ storage; each novel row requires $O(D)$ additional storage for its prototype and text anchor. Once its support images have been encoded, constructing a novel prototype requires $O(kD)$ arithmetic and no optimization state.

Scoring K candidates requires $O(KD)$ arithmetic in addition to the fixed reference calculations. Image encoding is shared across the vocabulary, and existing row scores can be reused when appending candidates for the same query.

5. Experiments

5.1. Data, access conditions, and implementation

We evaluate expansion on CIFAR-100 [28] with long-tailed base supervision. The training class counts follow an exponential schedule with imbalance parameter $\rho = 100$ and nominal maximum 500 images. Five images per class are reserved before constructing the long-tailed pool, so the largest available pool contains 495 images. The remaining long-tailed pool contains 10,894 images, split into 10,487 fitting images and 407 held-out validation images. Validation reserves up to five examples per class, with smaller allocations for the rarest classes.

For seeds $\{0, \ldots, 9\}$, stratified sampling within head, medium, and tail frequency buckets yields distinct partitions of 40 base, 40 novel, and 20 unused classes. Proportional quotas use largest-remainder rounding after shuffling class membership. Evaluation uses the 80 base and novel classes.

Oxford-IIIT Pets [29] provides a complementary fine-grained task with 37 cat and dog breeds. We use its official trainval/test split and reserve five training images per breed as support. From the remaining training images, we construct an exponential long-tailed pool with $\rho = 10$ and maximum class count 90, giving 1,168 fitting and 146 validation images. Each true partition contains 15 base, 15 novel, and seven unused breeds, sampled with the same allocation rule. The official test split contains 3,669 images. Unlike CIFAR, its images have varied spatial resolutions; RGB images are resized to 224×224 by PIL bicubic interpolation before the same CLIP normalization. These features are freshly extracted for this study.

All primary expansion readouts use frozen OpenAI CLIP ViT-L/14 or ViT-B/32 features from a single 224×224 bicubic-resized image with CLIP channel normalization. Fitting, validation, and support pools are disjoint. Supplementary Section C gives feature-cache and image-pool verification details.

Base ridge regularization is selected from $\{0.01, 0.1, 1, 10\}$ by base validation macro accuracy; the pseudo-base classifier uses the same rule. The primary comparison uses all five reserved support images, ten distinct class partitions and two arrival orders per partition. Earlier one/two/five-shot and repeated-calibration studies use three original partitions and are reported separately in Supplementary Section G.

Classifier fitting, regularization selection, and gate calibration use base classes exclusively. Five-shot CIFAR expansion consumes 200 support images for 40 novel classes. Its balanced test set supplies 4,000 base and 4,000 novel images for the expanded metrics.

ImageNet-R [30] extends the active vocabulary to 200 rendition classes: 100 base classes followed by five arrivals of 20 classes. After removing 59 byte-identical duplicates, per-class seeded splits reserve approximately 20% each for validation and test, then five support images. The fitting pool is downsampled by class rank with a nominal 100:1 schedule, yielding 2–136 images per class after availability caps. The resulting pools contain 6,438 fitting, 5,911 validation, 1,000 support and 5,911 test images. Ten frequency-stratified class partitions and two arrival orders use the same selection and reporting rules as the other datasets. RGB images are resized to 224×224 with bicubic interpolation and encoded by the same selected frozen checkpoint within each encoder comparison.

5.2. Compared readouts and statistical unit

Every readout shares its encoder, fitting pool, class partition and novel support. The retained table combines text and trained columns for base classes with text and support prototypes for novel classes. Fixed-gate removal sets the learned coefficient to zero; recalibrated removal independently reselects its remaining mixture. Dense anchored text–prototype fusion supplies visual means for every class. Tip-Adapter [8] supplies dense exponential semantic-cache retrieval from all permitted base fitting images and novel support. Both its published summed retrieval and an explicit class-mean control for unequal cache counts use the previously specified base-only selection grids. Recent FeNeC readouts [24] remain a three-partition controlled comparison in Supplementary Section F; their published-protocol numbers are not mixed with the expanded comparison.

Two simpler retained-classifier controls isolate normalization. Raw fusion uses cosine text/prototype scores and unnormalized ridge outputs. Fixed-temperature fusion divides each branch by one scalar, the mean per-query standard deviation across its fixed base reference columns on base-validation features. Scales are estimated once for the pseudo episode and once for the full base classifier, floored at 10^{-6}, and remain fixed throughout arrivals. These controls use the same coefficient/offset grids and pseudo-episode objective as the retained table. All three constructions can preserve existing scores; their recognition outcomes test the benefit of per-query reference statistics.

Two arrival orders use seeds $1000 + p$ and $1100 + p$ for partition p. Orders are averaged within each partition. We report means and sample standard deviations over ten distinct class assignments; the original three assignments are retained inside this expanded set. Paired differences use identical partition/order/session cells and, for constrained comparisons, their jointly feasible intersection. Percentile bootstrap intervals use 20,000 resamples of the paired partition means with seed 2026. These conditional intervals describe class-partition variation on shared fixed image pools and encoders, not independent-image or pretraining uncertainty. Supplementary Section I gives the full protocol and tables.

5.3. Validation-selected admission and achieved retention

For retention ratio r, each readout receives the same absolute target $\tau_r = rA^{\text{closed}}_{B,\text{ref}}$, using the retained table's closed-base accuracy on validation. We select the largest feasible group offset from the margins of correctly classified base-validation queries, then apply it to test queries without actual novel validation or test labels. A target above a readout's within-base ceiling is infeasible. We report both test group accuracies and separately count whether $A_{B,\text{test}} \geq rA^{\text{closed}}_{B,\text{ref,test}}$. That test criterion evaluates the fixed policy; it never selects its threshold.

On ImageNet-R, all ten partitions are validation-feasible for the retained, dense-prototype and class-mean cache readouts. Their novel test accuracies are 84.63%, 83.94% and 81.61%, with base test accuracies 86.15%, 86.01% and 86.08%, respectively (Table 1). The retained policy attains the separate 95% test target in 2 of ten partitions. Validation feasibility is therefore reported independently of realized test retention. The smaller-query study in Supplementary Section H measures threshold-label demand with the original fusion calibration held fixed.

Table 1: Validation-selected 95% retention policy with ViT-L/14 after five arrivals. Accuracies are percentages, mean ± sample standard deviation over feasible class partitions. Feasible counts validation-feasible partitions; Test met counts those attaining the separate test-retention criterion. All readouts share the same ten assignments and per-partition target; available means can cover different feasible subsets. The full ViT-B/32 and eight-readout comparisons appear in Supplementary Section I.

Data	Readout	Base	Novel	Feasible	Test met
CIFAR-100	Retained references	84.26 ± 1.91	77.10 ± 2.53	10/10	7/10
CIFAR-100	Recalibrated removal	82.91 ± 1.62	66.03 ± 13.04	6/10	1/6
CIFAR-100	Dense text + prototypes	84.00 ± 1.76	75.10 ± 7.15	8/10	4/8
CIFAR-100	Tip-Adapter mean	83.19 ± 1.72	68.69 ± 10.39	9/10	2/9
CIFAR-100	Raw-score fusion	82.11 ± 2.51	64.45 ± 13.35	7/10	1/7
CIFAR-100	Fixed temperatures	83.72 ± 2.24	70.93 ± 10.56	10/10	5/10
Pets	Retained references	89.95 ± 2.67	93.79 ± 3.83	10/10	1/10
Pets	Recalibrated removal	90.48 ± 3.04	93.44 ± 3.56	8/10	3/8
Pets	Dense text + prototypes	89.46 ± 3.08	94.47 ± 2.60	10/10	2/10
Pets	Tip-Adapter mean	90.73 ± 2.17	91.39 ± 4.32	10/10	4/10
Pets	Raw-score fusion	91.35 ± 2.91	92.72 ± 2.70	9/10	4/9
Pets	Fixed temperatures	90.48 ± 3.42	94.21 ± 3.19	9/10	3/9
ImageNet-R	Retained references	86.15 ± 0.94	84.63 ± 0.95	10/10	2/10
ImageNet-R	Recalibrated removal	86.16 ± 0.71	66.88 ± 22.79	10/10	3/10
ImageNet-R	Dense text + prototypes	86.01 ± 0.94	83.94 ± 1.10	10/10	3/10
ImageNet-R	Tip-Adapter mean	86.08 ± 1.08	81.61 ± 1.51	10/10	4/10
ImageNet-R	Raw-score fusion	86.15 ± 0.68	65.55 ± 20.42	10/10	2/10
ImageNet-R	Fixed temperatures	85.99 ± 0.71	82.92 ± 1.03	10/10	1/10

5.4. Classifier value over successive arrivals

Five sessions add eight CIFAR classes, three Pets breeds or 20 ImageNet-R classes at each arrival. With the base-only selected offset fixed throughout, $\overline{A} = 5^{-1} \sum_{t=1}^{5} A_{\mathrm{all},t}$ is average incremental macro accuracy; final accuracy uses the complete active vocabulary. Table 2 reports paired retained-minus-comparator differences, separating recalibration after removal from replacement by dense prototypes or semantic caches.

Table 2: Paired classifier value with fixed base-only calibrated gates and offsets. Entries are mean gains in percentage points [conditional paired 95% bootstrap CI] across ten class partitions after averaging the two arrival orders. Positive values favor retention. The interval resamples class assignments on the fixed data; it is not a population-performance guarantee.

Encoder	Data	Comparator	Paired avg. gain [95% CI]	Paired final gain [95% CI]
L/14	CIFAR-100	Recalibrated removal	3.00 [2.23, 3.75]	2.62 [1.81, 3.43]
L/14	CIFAR-100	Dense text + prototypes	0.15 [-0.46, 0.64]	-0.21 [-0.85, 0.30]
L/14	CIFAR-100	Tip-Adapter mean	1.44 [0.68, 2.10]	1.24 [0.41, 2.01]
L/14	Pets	Recalibrated removal	-0.14 [-1.53, 0.86]	-0.78 [-2.84, 0.60]
L/14	Pets	Dense text + prototypes	-0.98 [-2.52, 0.22]	-1.54 [-4.08, 0.35]
L/14	Pets	Tip-Adapter mean	-0.06 [-1.76, 1.41]	-0.39 [-2.92, 1.71]
L/14	ImageNet-R	Recalibrated removal	2.30 [1.95, 2.59]	1.71 [1.28, 2.06]
L/14	ImageNet-R	Dense text + prototypes	0.71 [0.46, 0.92]	0.41 [0.03, 0.72]
L/14	ImageNet-R	Tip-Adapter mean	1.42 [1.15, 1.62]	1.13 [0.76, 1.43]
B/32	CIFAR-100	Recalibrated removal	2.38 [1.44, 3.34]	1.80 [1.01, 2.65]
B/32	CIFAR-100	Dense text + prototypes	0.86 [-0.15, 1.76]	0.43 [-0.47, 1.24]
B/32	CIFAR-100	Tip-Adapter mean	0.96 [-0.04, 1.84]	0.51 [-0.34, 1.33]
B/32	Pets	Recalibrated removal	-0.44 [-1.64, 0.81]	-2.00 [-3.80, -0.19]
B/32	Pets	Dense text + prototypes	-0.39 [-1.85, 1.02]	-1.17 [-3.22, 0.67]
B/32	Pets	Tip-Adapter mean	-0.74 [-1.62, 0.12]	-1.85 [-3.40, -0.59]
B/32	ImageNet-R	Recalibrated removal	3.07 [2.82, 3.34]	2.39 [2.19, 2.61]
B/32	ImageNet-R	Dense text + prototypes	0.90 [0.69, 1.11]	0.46 [0.26, 0.66]
B/32	ImageNet-R	Tip-Adapter mean	1.71 [1.44, 1.98]	1.34 [1.16, 1.52]

On ImageNet-R, the ViT-L/14 retained table reaches 87.24% average incremental accuracy. Its paired gains are 0.71 [0.46, 0.92] points over dense prototypes and 1.42 [1.15, 1.62] over class-mean cache retrieval. With ViT-B/32, these gains are 0.90 [0.69, 1.11] and 1.71 [1.44, 1.98] points. The ImageNet-R gain therefore transfers across the two encoder architectures under identical image-pool access. CIFAR gains over classifier-free recalibration also appear with both encoders, while the dense-prototype comparison has intervals spanning zero. On

Pets, average incremental intervals span zero for all three comparators with both encoders; the final B/32 comparisons can favor the alternatives. These regimes distinguish the value of stored learned evidence from a claim of universal accuracy superiority.

5.5. Matched-test-retention capacity and simple fusion

The descriptive frontier fixes each mixture and asks

$$N_r = \max_d A_N(d) \quad \text{subject to} \quad A_B(d) \geq r A_{B,\text{ref,test}}^{\text{closed}}. \tag{15}$$

This sweep uses base test labels to characterize attainable capacity; it is separate from the validation-selected policy above. Sorting correctly classified base-query margins with macro weights identifies the largest feasible offset; Supplementary Section D derives its exact empirical envelope. We evaluate $r \in \{0.80, 0.85, 0.90, 0.95, 0.99\}$, with the same per-partition reference target for every readout. Infeasible targets are missing with explicit counts.

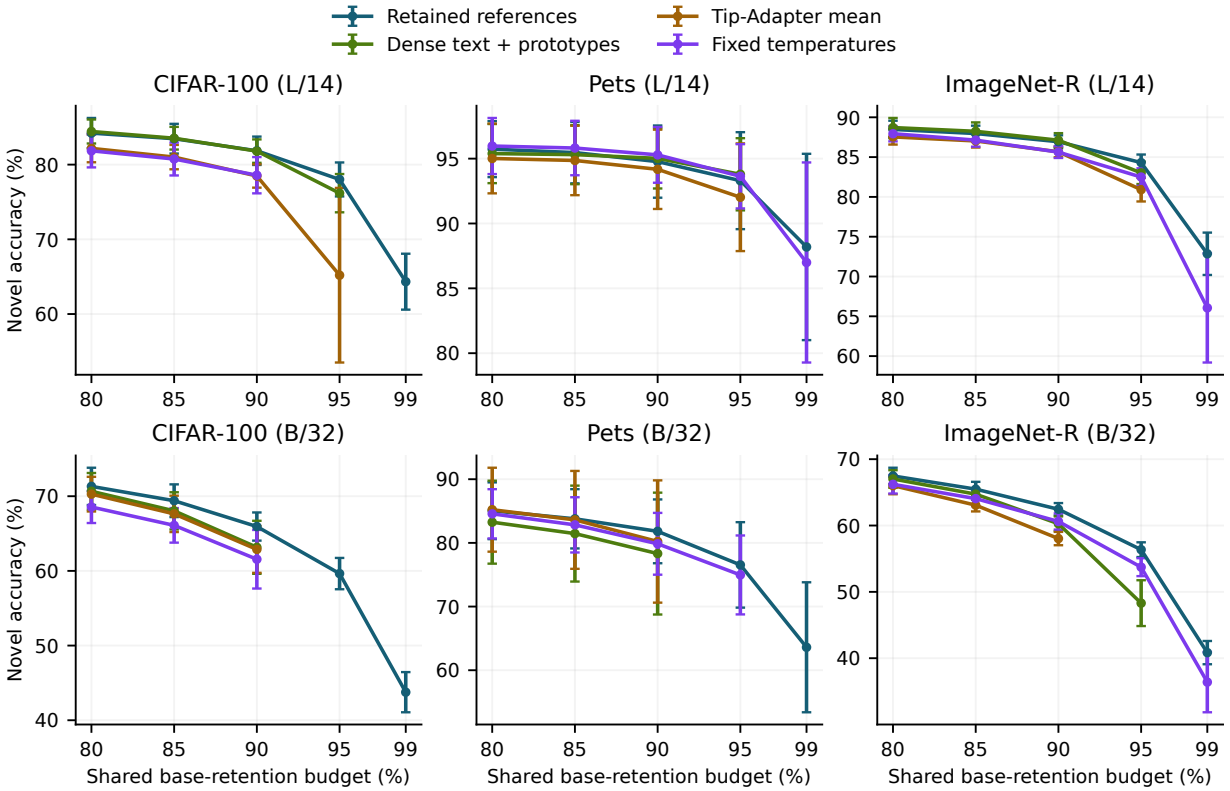


Figure 2: Descriptive novel-recognition frontiers over ten class partitions with two frozen CLIP encoders. Every readout receives the retained model's common absolute target within a partition and encoder. Error bars are partition-level sample standard deviations after averaging arrival orders. A point is drawn only when all ten partitions are feasible; Supplementary Section I reports available partial sets and counts. Thresholds use test labels for this capacity analysis and are not deployment calibration.

Table 3: Normalization controls with ViT-L/14. Fixed-gate average covers five arrivals; matched base/novel accuracy uses the descriptive 95% test-retention budget after the fifth arrival. Values are percentages, mean ± partition-level sample standard deviation; counts identify feasible sets. All fusion controls share the pseudo-episode grids and retained base classifier.

Data	Fusion	Fixed-gate avg.	Matched base	Matched novel	Feasible
CIFAR-100	Retained references	83.03 ± 1.19	83.91 ± 1.42	78.02 ± 2.27	10/10
CIFAR-100	Raw-score fusion	79.70 ± 2.64	83.25 ± 1.17	62.37 ± 6.38	6/10
CIFAR-100	Fixed temperatures	81.56 ± 0.98	83.64 ± 1.21	74.53 ± 2.70	9/10
Pets	Retained references	92.23 ± 2.87	91.56 ± 1.87	93.30 ± 3.74	10/10
Pets	Raw-score fusion	87.94 ± 14.58	91.57 ± 1.88	92.39 ± 2.75	10/10
Pets	Fixed temperatures	89.39 ± 10.15	91.56 ± 1.87	93.64 ± 2.48	10/10
ImageNet-R	Retained references	87.24 ± 0.59	86.42 ± 0.53	84.31 ± 1.01	10/10
ImageNet-R	Raw-score fusion	83.70 ± 2.58	86.42 ± 0.56	70.85 ± 8.03	9/10
ImageNet-R	Fixed temperatures	86.30 ± 0.38	86.43 ± 0.53	82.49 ± 1.11	10/10

At the ImageNet-R 95% budget, the retained ViT-L/14 readout reaches 86.42% base and 84.31% novel accuracy. Relative to fixed-temperature fusion, its paired average incremental gain is 0.94 [0.74, 1.12] points. This comparison isolates per-query branch scaling from the append-only invariant shared by both constructions. The complete budget curves and simple controls show where recognition value accompanies score preservation, while the Pets comparison separates structural preservation from an accuracy advantage. The imbalance schedules are held fixed in these experiments; differences across datasets and encoders describe the observed regimes rather than a causal effect of changing the imbalance ratio.

6. Discussion

Retained discrimination has an admission value beyond its closed-base accuracy. Under a shared base requirement, a classifier that separates the original classes more effectively can admit novel candidates at a more permissive threshold. The matched frontiers express this value directly as novel accuracy, while their feasible regions identify the base service levels that each readout can support.

Sequential comparisons make the contribution measurable across repeated arrivals. Fixed-gate removal evaluates the learned branch inside a selected mixture; classifier-free recalibration measures the value that remains after restoring the mixture's operating point; dense base

prototypes provide a stronger replacement alternative. These distinctions connect retained parameters to recognition outcomes rather than treating parameter preservation as the final objective.

Reference anchoring supplies a consistent score frame for this evaluation. Stored base columns and reference statistics remain fixed, branch coefficients transfer from base-only episodes, and the group threshold controls novel admission. A fixed threshold gives append-only score preservation; validation-based threshold adjustment supplies an explicit retention policy. Both operate on the same stored classifier.

7. Conclusion

We presented retention-constrained class expansion with reference-anchored candidate tables. The formulation measures novel recognition at a common base-accuracy requirement and quantifies the retained classifier's paired contribution across successive arrivals. Retention frontiers quantify the operating regimes in which learned base discrimination supports novel admission; five-session comparisons separate this contribution from calibration and prototype replacement. Preserving the original classifier becomes a measurable source of recognition capacity within a consistent expandable score system.

CRediT authorship contribution statement

Junyi Zhang: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft, Writing – review and editing.

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Declaration of competing interest

The author declares that there are no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The experiments use public CIFAR-100, Oxford-IIIT Pets and ImageNet-R data and the publicly released CLIP ViT-L/14 checkpoint. Code, source-hashed experimental records, prediction archives, manuscript sources, and table/figure generation scripts are available at `https://github.com/Cheerssom/Reference-Anchored-Candidate-Tables`. Dataset images, pretrained weights, and feature caches are obtained separately using the repository instructions.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

OpenAI Codex was used to assist with literature discovery, manuscript drafting and revision, experiment and verification scripts, and LaTeX and table preparation. The author is responsible for the accuracy, interpretation and originality of the submitted work and for reviewing and approving the final manuscript and supporting materials.

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Supplementary material: Retention-constrained class expansion with reference-anchored candidate tables

Junyi Zhang

Sections A–H retain the original three-partition ViT-L/14 auxiliary and validation studies. Section I reports the expanded ten-partition, two-encoder primary comparison. Their statistical units are stated separately.

Appendix A. Auxiliary readout and mechanism analyses

For the auxiliary readout and support-budget comparisons, the gate grid is $a, u \in \{0, 0.25, 0.5, 1, 2\}$ and $d \in \{-2, -1.5, \ldots, 2\}$. Offset-calibrated controls use their separately stated grids. The main retention/value evaluation uses the common expanded grid in Section 4.5. All experiments share the same frozen representations, true class partitions and disjoint support pools.

Appendix A.1. Readouts and evaluation measures

All compared readouts use the same CLIP image embeddings, text anchors, class partitions, and novel support images. The text baseline uses cosine similarity to class names. The visual prototype baseline uses means of base fitting examples and means of novel support examples. Text–prototype fusion adds the two cosine branches with a mixing coefficient selected on the same pseudo-novel calibration episode; its grid is $\{0, 0.25, 0.5, 1, 2, 4\}$, including pure text.

Tip-Adapter uses a balanced k-image cache per candidate and scores $100q_T + \alpha q_{\text{cache}}$, with affinity $\exp[\beta(v^\top f(x_i) - 1)]$. The same base-only episode selects $\alpha \in \{0, 0.5, 1, 2, 4\}$ and $\beta \in \{1, 2, 5, 10\}$. Cache images come from the reserved pool for both groups; base-trained readouts use the full base fitting set.

We also compare raw evidence fusion and candidate-normalized fusion. Raw fusion uses unstandardized text, learned, and prototype scores with the same coefficient grid as the proposed table. Candidate-normalized fusion standardizes text over all active classes, learned scores over the base classes, and prototypes over active novel classes. It shares the

Table S1: Five-shot expansion from 40 base classes to 80 classes. All values are accuracy percentages, reported as mean ± sample standard deviation over three distinct class partitions. All readouts share frozen CLIP features; coefficients are selected using base-only calibration. Base and novel performance jointly specify each operating point.

Readout	Base	Novel	All	Harmonic
CLIP text	78.32 ± 0.25	76.86 ± 1.97	77.59 ± 0.86	77.57 ± 0.88
Visual prototypes	77.53 ± 2.03	60.92 ± 3.94	69.23 ± 2.09	68.17 ± 2.49
Text + prototypes	82.47 ± 0.87	77.53 ± 0.47	80.00 ± 0.56	79.92 ± 0.55
Tip-Adapter (tuned)	78.38 ± 0.24	77.42 ± 1.72	77.90 ± 0.83	77.89 ± 0.83
Raw evidence fusion	78.32 ± 0.25	76.86 ± 1.97	77.59 ± 0.86	77.57 ± 0.88
Candidate-normalized table	85.36 ± 0.66	70.83 ± 4.33	78.09 ± 2.02	77.36 ± 2.48
Reference-anchored table	86.03 ± 1.40	66.90 ± 4.23	76.46 ± 1.43	75.19 ± 2.14

base classifier and calibration protocol with reference-anchored fusion. This comparison changes the normalization population while holding the principal evidence sources fixed.

We report macro base accuracy A_B, macro novel accuracy A_N, all-class macro accuracy, and the harmonic mean in Eq. (14). Both datasets use equal numbers of base and novel classes, so all-class macro accuracy is $(A_B + A_N)/2$. On the balanced CIFAR test set this also equals sample accuracy; on Pets the macro measure gives each breed equal weight. Stability measures include maximum absolute change in an existing score, change of the maximizing class restricted to base classes, and prediction changes over the expanded vocabulary. Error flips are reported as fractions of all base test images, not conditional fractions of previously correct images.

Appendix A.2. Recognition after five-shot class expansion

Reference-anchored fusion reaches 86.03 ± 1.40 base accuracy and 66.90 ± 4.23 novel accuracy (Table S1). Its base accuracy is 7.71 percentage points above text-only classification and 3.56 points above text–prototype fusion, providing a retention-oriented operating point for class expansion.

Text–prototype fusion uses visual evidence for every class and reaches 80.00 ± 0.56 all-class accuracy and 77.53 ± 0.47 novel accuracy. Candidate-normalized fusion reaches 85.36 ± 0.66 base and 70.83 ± 4.33 novel accuracy. These points characterize the recognition trade-off accompanying the reference-anchored score system.

Raw fusion selects the text-only point in all five-shot CIFAR partitions. Both normalization rules activate learned evidence in their selected solutions, demonstrating the role of scale alignment when combining classifier logits with cosine similarities.

Table S2: Five-shot Oxford-IIIT Pets expansion from 15 to 30 breeds under long-tailed base supervision. Values are macro accuracy percentages, mean ± sample standard deviation across three distinct breed partitions. Grids and the base-only calibration rule are transferred from the CIFAR experiment without dataset-specific search expansion.

Readout	Base	Novel	All	Harmonic
CLIP text	92.87 ± 3.37	94.05 ± 1.35	93.46 ± 1.59	93.43 ± 1.63
Visual prototypes	92.08 ± 2.61	68.30 ± 4.62	80.19 ± 2.33	78.35 ± 2.97
Text + prototypes	94.77 ± 2.91	93.91 ± 1.55	94.34 ± 1.43	94.31 ± 1.44
Tip-Adapter (tuned)	93.56 ± 2.45	94.07 ± 1.18	93.81 ± 1.27	93.80 ± 1.28
Raw evidence fusion	90.13 ± 8.01	91.18 ± 4.27	90.66 ± 6.13	90.63 ± 6.18
Candidate-normalized table	96.85 ± 1.50	86.49 ± 7.64	91.67 ± 4.56	91.29 ± 4.99
Reference-anchored table	97.18 ± 2.25	87.79 ± 6.18	92.49 ± 2.83	92.14 ± 3.23

Table S3: Additional five-shot controls with the same frozen encoder, support images, and base-only calibration. Values are mean ± sample standard deviation across three partitions. The random-projection readout is a controlled frozen RanPAC-style adaptation described in the text.

CIFAR-100

Readout	Base	Novel	All
Text + group offset	78.93 ± 2.75	75.56 ± 4.40	77.24 ± 0.89
Anchored text + prototypes	82.40 ± 2.02	76.19 ± 3.18	79.30 ± 0.59
Frozen RP-ridge (2000)	70.18 ± 0.69	52.52 ± 4.27	61.35 ± 1.83

Oxford-IIIT Pets

Readout	Base	Novel	All
Text + group offset	95.29 ± 3.80	89.85 ± 4.86	92.57 ± 2.31
Anchored text + prototypes	94.58 ± 3.22	90.38 ± 5.41	92.48 ± 1.10
Frozen RP-ridge (2000)	93.49 ± 3.32	73.70 ± 5.13	83.60 ± 1.13

Appendix A.3. Fine-grained expansion on Oxford-IIIT Pets

On fine-grained Pets expansion, reference anchoring reaches 97.18 ± 2.25 base and 87.79 ± 6.18 novel accuracy (Table S2). Base accuracy is 2.41 points above text–prototype fusion. Mean base/novel accuracy also increases from 96.85%/86.49% with candidate normalization to 97.18%/87.79% with reference anchoring.

Text–prototype fusion reaches 94.34 ± 1.43 all-class macro accuracy; the reference-anchored table reaches 92.49 ± 2.83 with greater retention of base-breed discrimination. Both datasets use the same search grids and base-only calibration rule.

Appendix A.4. Calibration controls and analytic incremental readouts

Offset-calibrated controls share the same pseudo-novel episode (Table S3). Text + group offset uses base-referenced text scores and a novel offset. Anchored text + prototypes uses fixed base references for both branches and visual evidence for every class. Harmonic selection searches offsets from -4 to 4 in steps of 0.25 and prototype weights $\{0, 0.25, 0.5, 1, 2, 4\}$. Fixed-reference text–prototype fusion also satisfies Proposition 1; the score invariant applies across availability masks.

Anchored text–prototype fusion attains 82.40% base and 76.19% novel accuracy on CIFAR, and 94.58%/90.38% on Pets. Offset-calibrated text attains 78.92%/75.56% and

Table S4: Validation retention-targeted calibration at five shots. The pseudo-base target is 95% of the closed reference-table pseudo-base accuracy. Among feasible grid points, selection maximizes pseudo-novel accuracy; infeasible partitions use the point with highest pseudo-base accuracy, then pseudo-novel accuracy. Feasible counts refer to validation partitions. Test values are percentages, mean ± sample standard deviation; test retention is not constrained.

Data	Readout	Base	Novel	All	Feasible
CIFAR	Text + group offset	81.47 ± 1.13	71.49 ± 5.03	76.48 ± 1.99	1/3
CIFAR	Anchored text + prototypes	82.84 ± 1.50	77.11 ± 4.46	79.98 ± 1.61	3/3
CIFAR	Candidate-normalized table	84.69 ± 1.92	72.45 ± 4.14	78.57 ± 1.54	3/3
CIFAR	Reference-anchored table	84.56 ± 1.31	73.88 ± 5.01	79.22 ± 2.21	3/3
Pets	Text + group offset	95.29 ± 3.80	89.85 ± 4.86	92.57 ± 2.31	3/3
Pets	Anchored text + prototypes	94.03 ± 3.37	90.96 ± 5.91	92.50 ± 1.27	3/3
Pets	Candidate-normalized table	97.25 ± 1.78	85.26 ± 7.25	91.26 ± 4.40	3/3
Pets	Reference-anchored table	97.18 ± 2.25	87.79 ± 6.18	92.49 ± 2.83	3/3

95.29%/89.85%, respectively. Together with the heterogeneous table, these controls separate group calibration, evidence configuration, and the shared fixed-reference property.

The analytic baseline follows the random-projection phase of RanPAC [1]. A fixed Gaussian matrix maps CLIP features to 2000 ReLU coordinates, and $(G + \lambda I)W = Q$ accumulates Gram matrices and class cross-products from base fitting data and novel support. Base validation macro accuracy selects $\lambda \in \{10^{-8}, \ldots, 10^{8}\}$; the pseudo episode selects a novel offset. This frozen adaptation uses base validation in place of the original within-task MSE selection. It attains 70.18%/52.53% base/novel accuracy on CIFAR and 93.49%/73.70% on Pets. Novel statistics refit the analytic readout; candidate-table expansion appends rows to its fixed readout.

Under the common 95% validation retention target, reference anchoring reaches 84.56% base and 73.87% novel test accuracy on CIFAR, compared with 82.84%/77.11% for anchored text–prototype fusion (Table S4). Both are feasible on all three validation partitions. On Pets, the corresponding test points are 97.18%/87.79% and 94.03%/90.96%. Text alone reaches the CIFAR validation target in one partition. Targets constrain validation selection; the table reports the resulting test operating points. Complete 95% and 99% results accompany the experimental records.

Appendix A.5. Vocabulary growth and score preservation

The vocabulary test appends nested sets of 10, 20, and 40 novel classes to the 40 CIFAR base classes, using sorted novel IDs and fixed five-shot gates. At each stage, we compare existing scores with the original base decision.

Reference anchoring gives zero measured base-score drift and zero base-only ranking

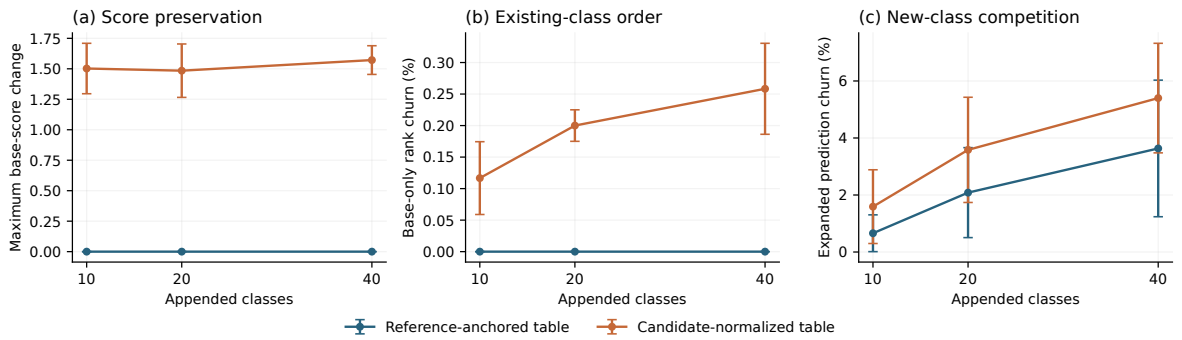


Figure S1: Vocabulary growth with five-shot novel prototypes. Each curve averages three true class partitions; error bars show sample standard deviation. Reference-anchored fusion preserves existing scores and base-only rankings. Expanded predictions can still change when a new candidate wins. Maximum score drift is measured across all test queries and existing base classes.

Table S5: Reference-anchored expansion across novel support budgets. Accuracy percentages are aggregated across partitions after averaging support subsets within each partition.

Shots	Base	Novel	All	Harmonic
1	84.57 ± 0.88	70.86 ± 5.41	77.72 ± 2.27	77.00 ± 2.85
2	85.65 ± 1.45	70.05 ± 1.13	77.85 ± 0.67	77.04 ± 0.67
5	86.03 ± 1.40	66.90 ± 4.23	76.46 ± 1.43	75.19 ± 2.14

churn in all nine CIFAR partition–vocabulary conditions (Fig. S1(a,b)). Candidate normalization changes existing scores by up to approximately 1.29–1.74 normalized units and changes a fraction of base-only predictions. The Pets test appends 3, 7, and 15 breeds and likewise records zero anchored score drift and ranking churn in all nine conditions.

At 40 appended classes, expanded predictions change on $3.63 \pm 2.40\%$ of base images, including $1.84 \pm 1.54\%$ correct-to-wrong changes (Fig. S1(c)). Wrong-to-correct changes are zero, as specified by Corollary 1. The preserved base scores isolate these changes as competition from appended rows.

Five-session expansion adds eight CIFAR classes or three Pets breeds per session, with two random arrival orders per partition. Across 30 session records per dataset, comparisons of all previously admitted rows, including novel rows, give maximum floating-point score differences of 2.85×10^{-5} on CIFAR and 3.58×10^{-6} on Pets. Both arrival orders produce identical final predictions in every partition under class-ID tie breaking.

Appendix A.6. Support budget and evidence availability

At one, two, and five shots, base accuracy remains between 84.57% and 86.03%, while novel accuracy ranges from 66.90% to 70.86% (Table S5; Fig. S2). Calibration includes zero

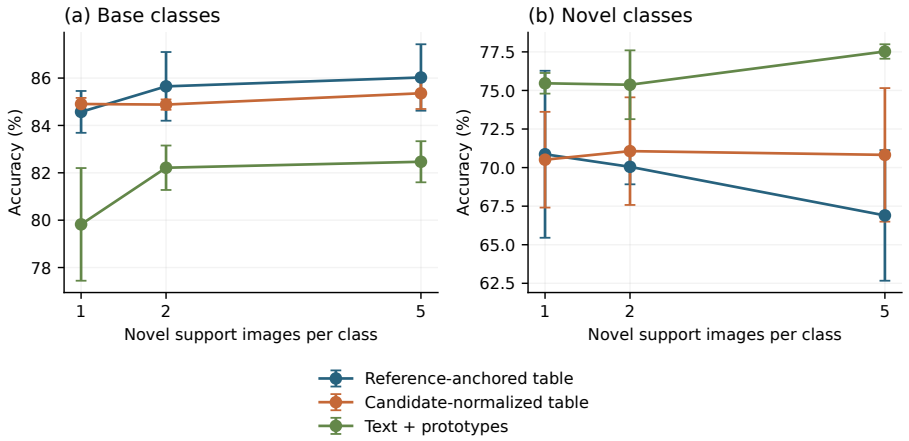


Figure S2: Base and novel accuracy at one, two, and five support images per novel class. The same fixed-reference scoring rule is used at every budget, with coefficients selected on the corresponding base-only pseudo episode. Error bars represent variation between class partitions.

Table S6: Evidence removal at the selected five-shot reference-anchored operating point, with all remaining coefficients fixed. Accuracy percentages are mean ± sample standard deviation over three class partitions.

Readout	Base	Novel	All	Harmonic
Reference-anchored table	86.03 ± 1.40	66.90 ± 4.23	76.46 ± 1.43	75.19 ± 2.14
Without learned evidence	58.10 ± 32.21	80.72 ± 4.47	69.41 ± 13.87	62.85 ± 25.08
Without novel prototypes	86.42 ± 1.71	47.17 ± 30.85	66.79 ± 14.62	56.15 ± 30.05
Without offset	86.14 ± 1.69	65.58 ± 7.35	75.86 ± 2.83	74.25 ± 4.08

branch coefficients, allowing the selected mixture to adapt to the support budget. Each budget defines a separately calibrated operating point: support determines the prototype evidence, and fusion determines its competition with learned base discrimination.

Appendix A.7. Mechanism ablation and operating trade-off

Removing learned evidence, novel prototypes, or the offset changes a different part of the selected decision rule (Table S6). With the remaining coefficients fixed, these interventions identify how each source contributes to the five-shot operating point. The independently calibrated text and text–prototype readouts provide the corresponding reduced-model comparisons.

Figure S3 sweeps d from -2 to 2 in steps of 0.5 with a and u fixed. Increasing d shifts the margin between the best base and novel candidates while preserving within-group rankings (Eq. (13)). The resulting curve identifies how the offset controls admission of new categories; a validation retention target selects the operating point.

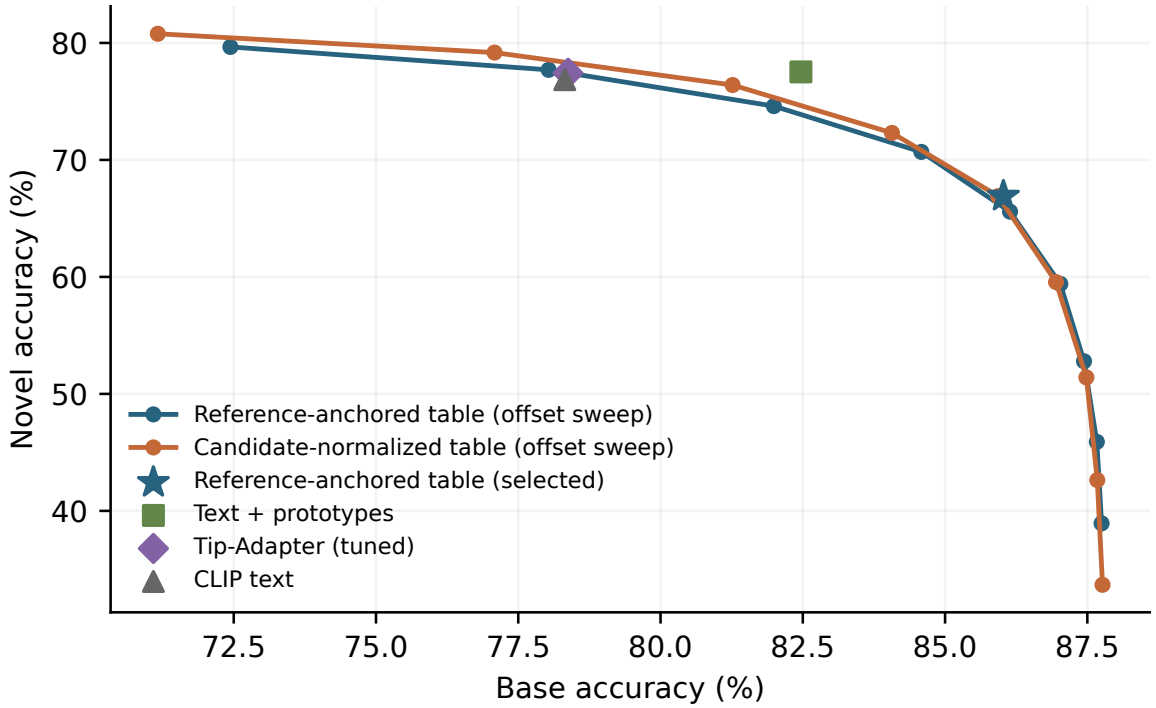


Figure S3: Base–novel operating curves for five-shot expansion. The offset is swept with the other selected coefficients fixed; these test curves are descriptive analyses and are not used to select the reported operating points. Symbols identify base-only calibrated points and matched baselines.

Appendix A.8. Readout efficiency

Candidate scoring from cached 768-dimensional features takes approximately 1.76, 0.75, and 0.78 ms per batch for 50, 60, and 80 candidates. Measurements use batches of 256 on an NVIDIA RTX 4070 Laptop GPU, five warm-up iterations, and 50 timed iterations with GPU synchronization. They include reference statistics and fusion and exclude image/text encoding. At these vocabulary sizes, launch and reduction overheads dominate the measured latency.

Appendix B. Validation-selected retention policy

Table S7 applies the common validation target after five arrivals. Raw records include all five target fractions and every arrival session. Infeasible targets are reported without substituting a zero novel accuracy. Comparisons with different feasible counts refer to different partition sets.

Table S7: Final-session test accuracies after the admission threshold is selected using base validation only. Budgets use the same closed-base validation reference for all methods within a partition. Values are percentages, mean ± sample standard deviation over feasible partitions; counts identify those partitions. Test accuracies are the outcomes of this operational policy and are not matched by test labels.

Data	Budget	Readout	Base	Novel	Feasible
CIFAR	90%	Retained classifier	78.33 ± 3.89	80.13 ± 2.25	3/3
CIFAR	90%	Classifier removed (fixed gates)	79.07 ± 3.94	70.79 ± 13.81	3/3
CIFAR	90%	No classifier (recalibrated)	78.73 ± 3.21	75.45 ± 5.43	3/3
CIFAR	90%	Anchored text + prototypes	80.18 ± 0.60	79.45 ± 0.76	3/3
CIFAR	95%	Retained classifier	82.99 ± 2.29	75.71 ± 3.14	3/3
CIFAR	95%	Classifier removed (fixed gates)	81.95 ± 0.04	71.54 ± 3.87	2/3
CIFAR	95%	No classifier (recalibrated)	81.85 ± 0.18	71.70 ± 4.10	2/3
CIFAR	95%	Anchored text + prototypes	81.95 ± 0.64	77.48 ± 1.52	2/3
CIFAR	99%	Retained classifier	86.48 ± 1.65	62.83 ± 10.11	3/3
CIFAR	99%	Classifier removed (fixed gates)	–	–	0/3
CIFAR	99%	No classifier (recalibrated)	–	–	0/3
CIFAR	99%	Anchored text + prototypes	86.85	53.85	1/3
Pets	90%	Retained classifier	84.87 ± 4.59	96.53 ± 0.78	3/3
Pets	90%	Classifier removed (fixed gates)	84.11 ± 2.76	96.44 ± 0.64	3/3
Pets	90%	No classifier (recalibrated)	82.00 ± 0.20	96.38 ± 0.52	3/3
Pets	90%	Anchored text + prototypes	85.69 ± 4.62	95.87 ± 1.57	3/3
Pets	95%	Retained classifier	90.29 ± 2.90	95.68 ± 0.83	3/3
Pets	95%	Classifier removed (fixed gates)	90.11 ± 1.17	95.71 ± 0.11	2/3
Pets	95%	No classifier (recalibrated)	89.88 ± 0.85	95.71 ± 0.11	2/3
Pets	95%	Anchored text + prototypes	89.37 ± 4.15	95.63 ± 1.55	3/3
Pets	99%	Retained classifier	96.43 ± 2.15	88.67 ± 7.89	3/3
Pets	99%	Classifier removed (fixed gates)	–	–	0/3
Pets	99%	No classifier (recalibrated)	–	–	0/3
Pets	99%	Anchored text + prototypes	97.70 ± 1.11	84.19 ± 12.46	2/3

Appendix C. Reproduction details

The expansion experiment is implemented in `src/experiments/fcs_expansion.py`. It can be run from any working directory. The manifest records the code hash, frozen-cache hash, image-pool hashes, class assignments, pseudo-class assignments, support indices, regularization choices, and selection grids. The frozen support features are computed with the same deterministic interpolation used for the frozen training cache. The cached image embeddings are checked numerically before scoring.

Every novel support image comes from the five-image-per-class reserve created before long-tail sampling. This supplies a consistent support budget even for tail classes whose fitting pool contains fewer than five images. At five shots the three subset seeds select the same images in different orders, so the reported standard deviation is calculated across the three distinct class partitions. One- and two-shot results average the three subsets

within each partition before computing the partition-level variation.

The main result JSON contains all compared readouts, including evidence-removal ablations. The vocabulary records contain the maximum score drift and both restricted and expanded prediction churn. The operating-curve records contain every swept offset, with no selection based on test outcomes. The efficiency records specify batch size and measured vocabulary size. The manuscript tables are generated directly from these files, and the summary CSV exposes both means and sample standard deviations.

The semantic prompts are "a photo of a {}.", "a bad photo of a {}.", "a cropped photo of a {}.", "a close-up photo of a {}.", "a bright photo of a {}.", "a good photo of a {}.", "a photo of the {}.", and a second occurrence of "a photo of a {}.". The repeated prompt follows the existing cached anchor construction and is retained consistently for all methods.

Frozen CLIP features are the sole image representation in the expansion experiments. Six fitting images are independently re-embedded and checked for cosine agreement above 0.999 with the cached features. Image hashes verify that fitting, validation, and reserved-support pools are disjoint. The fixed base text block is computed separately from novel text columns; sequential prototype products can exhibit the reported floating-point residuals when matrix widths change.

Appendix D. Interpreting group competition

Let $M_B(x) = \max_{b \in \mathcal{B}}[\widehat{q}_T(x, b) + a\widehat{q}_L(x, b)]$ and $M_N(x) = \max_{n \in \mathcal{N}}[\widehat{q}_T(x, n) + u\widehat{q}_P(x, n)]$. The classifier selects a novel class precisely when $M_N(x) + d > M_B(x)$, subject to the fixed tie-breaking rule. Hence, the offset is a threshold on the margin $M_B - M_N$. Increasing d cannot change the winning base class or the winning novel class when all other quantities are fixed. It changes which group supplies the final winner. This explains the direction of the operating curves without assuming a calibrated probability interpretation for the fused score.

For nested candidate sets and fixed references, adding novel rows can increase M_N but cannot change M_B. An existing-class example may therefore cross the admission boundary as the vocabulary grows, while its base scores remain identical. This distinguishes the deployment behavior measured by expanded prediction churn from the normalization behavior measured by base-only ranking churn.

Appendix D.1. Exact empirical retention envelope

For a fixed readout, let b_i and n_i be the canonical winning base and novel class for query i, and let $\Delta_i = M_B(x_i) - M_N(x_i)$. Define $g_i(d) = \mathbf{1}[d > \Delta_i \text{ or } (d = \Delta_i \text{ and } n_i < b_i)]$. With the class-macro weights w_i, the two group accuracies are

$$A_B(d) = \sum_{i:y_i \in \mathcal{B}} w_i \mathbf{1}[b_i = y_i](1 - g_i(d)), \tag{D.1}$$

$$A_N(d) = \sum_{i:y_i \in \mathcal{N}} w_i \mathbf{1}[n_i = y_i] g_i(d). \tag{D.2}$$

Thus A_B is nonincreasing and A_N is nondecreasing in d. For the stored finite-precision scores, an exact search need only consider each observed margin and its immediately preceding representable offset, together with the all-base and all-novel endpoints. Between adjacent margins every group decision is fixed. If τ exceeds $\sum_{i:y_i \in \mathcal{B}} w_i \mathbf{1}[b_i = y_i]$, the constraint is infeasible. Otherwise, sorting the margins of correctly classified base queries in descending order and accumulating their macro weights locates the boundary at which retained base mass first reaches τ. The tie rule selects that boundary or its predecessor. This is the largest feasible offset in the implementation's offset ordering. Monotonicity of A_N proves that it maximizes novel accuracy under the empirical constraint. The argument applies identically to base-validation selection; that policy's test outcome is evaluated on separate queries.

The envelope separates the within-base feasible ceiling, the admitted group threshold and the distribution of correctly classified novel-query margins. It therefore measures the joint operating value of a learned readout and its competition with arriving evidence, beyond the statement that stored scores are preserved.

Appendix E. Additional background and score-preservation proof

Appendix E.1. Long-tailed recognition and calibration

Long-tailed recognition addresses the bias induced by unequal class frequencies. Margin adjustment allocates different classification margins to frequent and rare classes [2]; class-balanced weighting uses effective sample counts [3]; and prior-aware objectives modify the relationship between logits and class probabilities [4, 5]. Representation–classifier decoupling provides a complementary route: a representation is learned first and a balanced classifier is then fitted to that representation [6]. These developments motivate our use of a balanced base classifier as one source of evidence.

Foundation models change the representation-learning budget and the role of task-specific adaptation. LIFT studies lightweight adaptation of pre-trained models for long-tail learning [7]. Low-rank updates provide an efficient mechanism for adapting large parameter matrices [8], including transformer backbones [9]. Our candidate interface can accept weights obtained through these routes. The expansion experiments here fix the encoder throughout and isolate the effect of the readout.

Long-tailed calibration studies confidence under class imbalance [10], while robust long-tailed learning addresses its interaction with label noise [11]. We study the reference population used to standardize evidence as candidate classes change. Class-balanced validation and separate base/novel metrics expose how the resulting score system allocates discrimination.

Appendix E.2. Semantic and visual evidence for few-shot recognition

CLIP constructs classifiers from natural-language supervision and supports classification by changing the text candidates [12]. CoOp learns prompt context for target tasks, while CoCoOp introduces conditional prompting to improve transfer across categories [13, 14]. These methods concern the construction or adaptation of semantic evidence. Our table keeps semantic anchors available to every class and combines them with evidence that may exist for only a subset of the vocabulary.

Prototype methods summarize class examples in an embedding space [15]. SimpleShot studies feature normalization for nearest-neighbor classification [16], and Tip-Adapter augments CLIP with a support-feature cache [17]. Few-shot transfer under distribution shift further motivates separating shared and task-specific knowledge [18]. Our comparisons use a shared encoder and novel support budget to examine evidence availability and its normalization reference.

Appendix E.3. Proof of the main score-preservation proposition

Proof. For an existing candidate c, each raw score $q_e(x, c)$ and mask m_{ec} is fixed. The means and variances in Eq. (9) are functions of x and $\mathcal{R}_e$, and neither depends on the appended rows. Therefore each normalized score in Eq. (10) is unchanged. The coefficients and offset of c are fixed, so their sum is unchanged. Subtracting the unchanged scores of any two existing rows gives an unchanged difference. A fixed tie-breaking rule preserves the maximizing existing row. □

The proposition covers both base rows and previously appended novel rows with fixed prototypes. The implementation computes the fixed base text block separately from novel columns to keep its numerical kernel independent of the appended width.

Corollary 1. *If every existing row has a preserved score, a changed final prediction after expansion must select an appended row. An example whose previous prediction was incorrect cannot become correct after adding classes when its true class belongs to the existing vocabulary.*

Proof. The best existing row is unchanged by Proposition 1. The expanded maximizer is either this row or a newly appended row. For an example labeled by an existing class, selecting a new row cannot correct a previous error, and keeping the existing maximizer leaves the previous error unchanged. □

Appendix F. Recent readout comparison and ImageNet-R extension

The extension protocol was recorded before ImageNet-R evaluation. Class partitions use seeds 0, 1 and 2, with exact frequency-stratified allocation of 100 base and 100 novel classes. Within each base group, seed $900 + p$ assigns 50 pseudo-base and 50 pseudo-novel classes. Base-only validation selects ridge regularization, branch mixtures and readout hyperparameters. No ImageNet-R test result selects these quantities. The retained table, removal controls and dense prototype replacement use the same grids as the CIFAR/Pets experiment.

FeNeC [19] is independently implemented from the paper's equations and checked against author-code commit `1575ea019fd583149c47c068f7f5ccc102ea4580`. Both variants use identity Tukey transformation for signed CLIP features and Euclidean k-means with a maximum of 100 iterations. The representative grid is $K \in \{1, 5, 25\}$ and the neighbor grid is $\{1, 3, 5\}$. FeNeC-M estimates one covariance per class, applies the author's ViT single-shrinkage and correlation normalization with $(\gamma_1, \gamma_2) \in \{(1, 1), (6.12, 8.10), (9.98, 0)\}$, then uses inverse squared-Mahalanobis-distance votes. FeNeC-E uses inverse unsquared-Euclidean-distance votes. Their native predictions are preserved when votes are converted to log votes, with a floor of 10^{-30}. The calibrated forms select a novel offset from $[-4, 4]$ in steps of 0.25 on the pseudo episode. Harmonic macro accuracy chooses the configuration.

The five-shot protocol caps a class's representative count at its sample count. A class with one fitting sample uses identity covariance, avoiding an undefined sample covariance. These are explicit sparse-data adaptations. A per-class independent summary can be cached before its arrival; the voting neighborhood always contains only currently admitted classes. Native and calibrated forms share each current-vocabulary score matrix, so their complete matched-retention frontiers coincide, while fixed-offset trajectories can differ. The final grids are documented with the code. FeNeC frontiers use its sparse global-neighbor votes, with unvoted classes at the common floor; feasibility refers to this output interface.

All ImageNet-R images are content-hashed before pool assignment. Fifty-nine byte-identical duplicates are removed globally. No retained content hash appears in two fitting, validation, support or test pools. The source archive has SHA-256 `18c6bf493b39a0d975d48e587437f562caab9c52ae6327dcfa9dd8eb54aa1b52`. The class-name mapping, image paths, achieved fitting counts, source hashes, calibrated settings and prediction margins accompany the code and records. These are task-specific splits of ImageNet-R, whose source release is a rendition benchmark [20].

Table S8: ImageNet-R matched-retention recognition at three budgets after five arrivals. Values are percentages, mean ± partition-level sample standard deviation over feasible partitions. Counts specify available partition sets.

Budget	Readout	Base	Novel	Feasible
90%	Retained classifier	81.83 ± 0.29	86.80 ± 0.27	3/3
90%	Classifier removed (fixed gates)	81.81 ± 0.31	85.53 ± 0.49	3/3
90%	No classifier (recalibrated)	81.81 ± 0.31	85.34 ± 0.46	3/3
90%	Anchored text + prototypes	81.83 ± 0.32	86.59 ± 0.58	3/3
90%	Tip-Adapter sum + offset	81.82 ± 0.30	83.74 ± 0.25	3/3
90%	Tip-Adapter mean + offset	81.83 ± 0.30	85.30 ± 0.23	3/3
95%	Retained classifier	86.38 ± 0.32	84.48 ± 0.72	3/3
95%	Classifier removed (fixed gates)	86.37 ± 0.31	72.62 ± 12.51	3/3
95%	No classifier (recalibrated)	86.37 ± 0.31	72.69 ± 12.57	3/3
95%	Anchored text + prototypes	86.37 ± 0.32	83.32 ± 1.40	3/3
95%	Tip-Adapter sum + offset	86.41 ± 0.30	70.93 ± 10.32	3/3
95%	Tip-Adapter mean + offset	86.38 ± 0.33	81.58 ± 1.72	3/3
99%	Retained classifier	90.02 ± 0.34	73.69 ± 1.50	3/3
99%	Classifier removed (fixed gates)	–	–	0/3
99%	No classifier (recalibrated)	–	–	0/3
99%	Anchored text + prototypes	89.98 ± 0.47	47.62 ± 13.36	2/3
99%	Tip-Adapter sum + offset	–	–	0/3
99%	Tip-Adapter mean + offset	–	–	0/3

Table S9: ImageNet-R test outcomes with the 95% base-validation-only threshold policy. Validation feasibility is reported separately from achieved test accuracy. Tip-Adapter uses its base-only selected calibrated cache settings, followed by validation-only threshold selection.

Readout	Base test	Novel test	Validation feasible
Retained classifier	86.01 ± 0.59	84.73 ± 0.71	3/3
Classifier removed (fixed gates)	86.38 ± 0.18	57.35 ± 38.51	3/3
No classifier (recalibrated)	86.48 ± 0.20	57.10 ± 38.30	3/3
Anchored text + prototypes	86.36 ± 0.47	83.26 ± 1.49	3/3
Tip-Adapter sum + offset	86.10 ± 0.46	74.08 ± 5.48	3/3
Tip-Adapter mean + offset	86.68 ± 0.23	80.75 ± 2.09	3/3

Table S10: FeNeC matched-test-retention operating points at strict budgets. Values are percentages, mean ± sample standard deviation over feasible partitions; missing values have zero feasible partitions. Native and offset-calibrated forms share a score matrix and hence the same complete offset-sweep frontier.

Data	Budget	Readout	Base	Novel	Feasible
CIFAR-100	95%	FeNeC-M + offset	–	–	0/3
CIFAR-100	95%	FeNeC-E + offset	–	–	0/3
CIFAR-100	99%	FeNeC-M + offset	–	–	0/3
CIFAR-100	99%	FeNeC-E + offset	–	–	0/3
Pets	95%	FeNeC-M + offset	–	–	0/3
Pets	95%	FeNeC-E + offset	94.23	69.46	1/3
Pets	99%	FeNeC-M + offset	–	–	0/3
Pets	99%	FeNeC-E + offset	–	–	0/3
ImageNet-R	95%	FeNeC-M + offset	–	–	0/3
ImageNet-R	95%	FeNeC-E + offset	–	–	0/3
ImageNet-R	99%	FeNeC-M + offset	–	–	0/3
ImageNet-R	99%	FeNeC-E + offset	–	–	0/3

Appendix G. Dense caches, repeated calibration and measured readout cost

Appendix G.1. Dense semantic-cache protocol

The training-free Tip-Adapter score follows the ECCV paper and the author implementation [17]: cosine similarity to labeled cache keys is transformed by $\exp[-\beta(1-\text{similarity})]$, summed into classes and added to 100 times text similarity with weight α. The author code is pinned at commit `d0e2d6f8c5feb8b6ce937b757810761f7155d4d5`. All permitted base fitting images and five reserved novel support images supply the keys. The sum variant retains the published aggregation. A separate class-mean variant divides each class's retrieval score by its key count, isolating the effect of unequal long-tailed cache sizes. Both reuse the same single-view frozen features and text anchors as the tables. This comparison transfers the published readout under controlled feature access; the original augmented-cache extraction and trainable Tip-Adapter-F are distinct settings.

On each original base-only pseudo episode, we search $\beta \in \{0.1, 1, 2, 5, 10, 20, 50\}$ and $\alpha \in \{0.1, 1, 5, 10, 20, 50, 100, 200\}$. The native form selects both parameters with zero offset. The calibrated form jointly selects these parameters and an offset from $[-40, 40]$ in steps of 2.5, using macro harmonic accuracy. This 33-point offset grid reflects the raw 100-scaled CLIP score system. No actual novel query or test label is used in selection. Both forms are reported because their selected cache settings can differ. All matched-retention targets are the original retained readout's partition-specific absolute targets.

Appendix G.2. Repeated calibration and support budgets

The 81 configurations span three datasets, three original class partitions, $k \in \{1, 2, 5\}$ and three joint calibration/support trials. Trial zero reuses the original pseudo split. The other trials select $\lfloor |\mathcal{B}|/2 \rfloor$ pseudo-base classes with seeds $1701 + p$ and $2701 + p$. Pseudo classifiers select ridge regularization on pseudo-base validation. Support-index seeds $7, 8, 9$ select one- and two-shot subsets from each five-image reserve. At five shots, the support set is identical across trials: variation comes from the pseudo split. These are correlated trials of the same underlying class partitions. We average the orders and trials within each partition and report the mean and sample standard deviation of the three partition means.

Each configuration independently calibrates the retained, recalibrated-removal, dense-prototype and dynamic-reference readouts on the same base-only episode. Fixed-gate removal preserves the retained mixture's remaining parameters. Dynamic normalization uses the current text candidates and current novel prototypes as references, with the same

Table S11: Complete five-shot fixed-gate readout comparison. Values are percentages, mean ± sample standard deviation over three class partitions after averaging the two arrival orders. The paired gain is retained minus comparator average incremental accuracy; a negative gain favors the comparator. Native and calibrated cache forms are selected separately.

Data	Readout	Incremental avg.	Final	Paired avg. gain
CIFAR-100	Retained classifier	82.02 ± 0.84	79.31 ± 1.14	0.00 ± 0.00
CIFAR-100	Classifier removed (fixed gates)	71.40 ± 14.78	68.03 ± 15.63	10.62 ± 15.22
CIFAR-100	No classifier (recalibrated)	79.84 ± 0.82	77.56 ± 1.02	2.18 ± 0.24
CIFAR-100	Anchored text + prototypes	81.56 ± 0.52	79.30 ± 0.59	0.46 ± 0.44
CIFAR-100	Tip-Adapter sum + offset	79.57 ± 0.41	77.21 ± 0.38	2.45 ± 0.63
CIFAR-100	Tip-Adapter mean + offset	81.74 ± 0.93	79.40 ± 1.11	0.28 ± 1.07
CIFAR-100	FeNeC-M + offset	66.60 ± 1.34	63.65 ± 1.55	15.42 ± 0.49
CIFAR-100	FeNeC-E + offset	72.63 ± 1.34	69.00 ± 1.97	9.39 ± 0.68
CIFAR-100	FeNeC-M (native)	59.11 ± 0.85	53.58 ± 1.27	22.91 ± 0.31
CIFAR-100	FeNeC-E (native)	72.63 ± 1.34	69.00 ± 1.97	9.39 ± 0.68
CIFAR-100	Tip-Adapter sum (native)	79.57 ± 0.41	77.21 ± 0.38	2.45 ± 0.63
CIFAR-100	Tip-Adapter mean (native)	81.74 ± 0.93	79.40 ± 1.11	0.28 ± 1.07
Pets	Retained classifier	94.09 ± 2.37	91.77 ± 3.16	0.00 ± 0.00
Pets	Classifier removed (fixed gates)	77.17 ± 26.01	76.85 ± 23.25	16.92 ± 28.33
Pets	No classifier (recalibrated)	94.20 ± 2.05	92.57 ± 2.31	-0.11 ± 0.56
Pets	Anchored text + prototypes	94.35 ± 0.60	92.48 ± 1.10	-0.26 ± 1.82
Pets	Tip-Adapter sum + offset	93.38 ± 2.32	91.50 ± 2.78	0.71 ± 1.21
Pets	Tip-Adapter mean + offset	94.47 ± 2.38	92.88 ± 2.51	-0.37 ± 1.61
Pets	FeNeC-M + offset	80.34 ± 1.29	75.66 ± 1.72	13.76 ± 1.61
Pets	FeNeC-E + offset	84.43 ± 2.37	79.82 ± 2.35	9.67 ± 1.50
Pets	FeNeC-M (native)	73.66 ± 4.73	66.74 ± 6.13	20.44 ± 2.69
Pets	FeNeC-E (native)	84.43 ± 2.37	79.82 ± 2.35	9.67 ± 1.50
Pets	Tip-Adapter sum (native)	94.35 ± 1.08	92.72 ± 1.65	-0.25 ± 1.32
Pets	Tip-Adapter mean (native)	95.15 ± 1.28	93.74 ± 1.29	-1.06 ± 1.33
ImageNet-R	Retained classifier	87.23 ± 0.21	85.40 ± 0.36	0.00 ± 0.00
ImageNet-R	Classifier removed (fixed gates)	77.13 ± 6.75	74.64 ± 7.19	10.09 ± 6.55
ImageNet-R	No classifier (recalibrated)	85.01 ± 0.57	83.65 ± 0.70	2.22 ± 0.40
ImageNet-R	Anchored text + prototypes	86.36 ± 0.24	84.67 ± 0.39	0.87 ± 0.29
ImageNet-R	Tip-Adapter sum + offset	84.62 ± 0.08	82.94 ± 0.07	2.61 ± 0.21
ImageNet-R	Tip-Adapter mean + offset	85.75 ± 0.30	84.19 ± 0.35	1.48 ± 0.35
ImageNet-R	FeNeC-M + offset	59.12 ± 0.54	55.79 ± 0.63	28.11 ± 0.46
ImageNet-R	FeNeC-E + offset	69.66 ± 0.97	66.09 ± 0.62	17.56 ± 0.80
ImageNet-R	FeNeC-M (native)	54.57 ± 0.47	49.71 ± 0.32	32.65 ± 0.37
ImageNet-R	FeNeC-E (native)	69.66 ± 0.97	66.09 ± 0.62	17.56 ± 0.80
ImageNet-R	Tip-Adapter sum (native)	84.62 ± 0.08	82.94 ± 0.07	2.61 ± 0.21
ImageNet-R	Tip-Adapter mean (native)	85.75 ± 0.30	84.19 ± 0.35	1.48 ± 0.35

learned branch and calibration grid as the retained table. Its own mixture is selected independently and stays fixed across arrivals. All retention targets are shared within each configuration. Feasibility counts refer to nine partition/trial combinations per support budget. When only some trials are feasible, the group accuracies average those trials within each partition before summarizing partitions; paired constrained comparisons use the intersection of feasible trial IDs.

Table S12: Calibration and support robustness. Average incremental and final accuracies use fixed gates. Paired gains are retained minus each independently calibrated comparator, after averaging two arrival orders and three trials within each partition. Values are percentages or percentage points, mean ± sample standard deviation across three partitions.

Data	Shots	Retained avg.	Δ recalibrated	Δ dense	Δ dynamic	Retained final
CIFAR-100	1	79.77 ± 0.61	0.55 ± 1.15	0.08 ± 0.72	0.10 ± 0.74	76.32 ± 1.22
CIFAR-100	2	80.20 ± 0.32	0.84 ± 0.67	-0.76 ± 0.52	0.55 ± 0.84	77.00 ± 0.85
CIFAR-100	5	80.57 ± 0.96	1.22 ± 1.18	-0.79 ± 1.08	0.62 ± 0.59	77.25 ± 1.09
Pets	1	93.94 ± 2.37	0.05 ± 0.67	-0.24 ± 0.48	-0.54 ± 0.11	91.60 ± 3.20
Pets	2	92.64 ± 1.82	-1.45 ± 1.19	-1.62 ± 1.15	-1.43 ± 1.09	90.09 ± 1.87
Pets	5	94.56 ± 1.74	0.78 ± 1.03	0.43 ± 0.80	0.47 ± 1.43	92.75 ± 2.00
ImageNet-R	1	86.16 ± 0.22	1.81 ± 0.33	1.01 ± 0.29	-0.07 ± 0.29	83.83 ± 0.14
ImageNet-R	2	86.58 ± 0.37	2.10 ± 0.28	0.62 ± 0.34	-0.07 ± 0.28	84.43 ± 0.67
ImageNet-R	5	87.16 ± 0.12	2.14 ± 0.37	0.54 ± 0.23	0.15 ± 0.19	85.36 ± 0.25

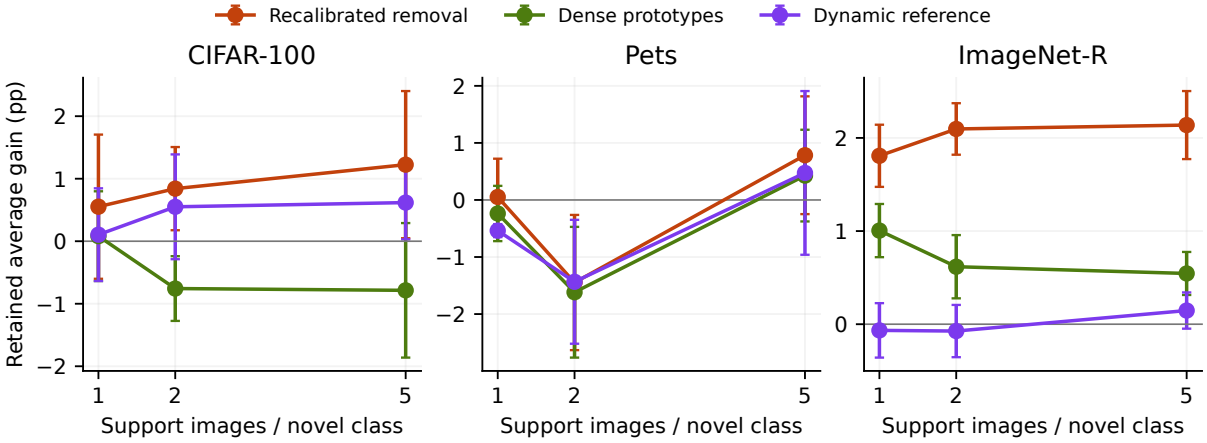


Figure S4: Paired average incremental gains under one-, two- and five-shot support. Error bars are partition-level sample standard deviations after averaging correlated calibration/support trials and arrival orders. The horizontal zero line identifies which readout has higher average accuracy in each setting.

Fixed references have zero restricted pre-offset candidate-ranking changes in all 648 append checks (81 configurations, two orders, four appends). The largest floating-point score differences are 1.40×10^{-4} on CIFAR, 5.36×10^{-6} on Pets and 2.19×10^{-5} on ImageNet-R. Dynamic references produce maximum score changes of 2.40, 1.78 and 1.30, respectively, and up to 429, 206 and 118 restricted prediction changes per append over

the full cached test query pool. These diagnostics compare the same previously active candidate IDs before adding the fixed group offset; they isolate changes induced by reference statistics. Recognition accuracy is reported separately in the fixed-gate and retention-policy tables.

Table S13: CIFAR-100: final-session outcomes in the repeated-calibration study at the 95% budget. Match uses base test labels descriptively; Val. uses base validation only and reports achieved test accuracy. Feasibility is counted over nine correlated partition/trial combinations.

Shots	Readout	Match base	Match novel	Feasible	Val. base	Val. novel	Feasible
1	Retained classifier	83.54 ± 0.81	73.03 ± 3.08	9/9	80.77 ± 2.34	76.30 ± 3.82	9/9
1	No classifier (recalibrated)	83.37 ± 0.57	59.17 ± 3.51	8/9	80.92 ± 1.45	72.28 ± 4.17	9/9
1	Anchored text + prototypes	83.37 ± 0.57	63.65 ± 3.03	8/9	80.38 ± 2.28	73.74 ± 5.16	9/9
1	Dynamic reference	83.54 ± 0.81	73.07 ± 3.81	9/9	81.10 ± 2.54	75.86 ± 4.56	9/9
2	Retained classifier	82.99 ± 0.46	73.84 ± 3.52	9/9	80.65 ± 1.93	76.40 ± 4.33	9/9
2	No classifier (recalibrated)	82.99 ± 0.46	61.20 ± 0.62	9/9	81.14 ± 1.01	71.44 ± 2.35	9/9
2	Anchored text + prototypes	82.99 ± 0.46	71.62 ± 6.74	9/9	80.81 ± 1.65	76.62 ± 4.51	9/9
2	Dynamic reference	82.99 ± 0.46	74.70 ± 3.56	9/9	80.37 ± 1.53	77.66 ± 2.64	9/9
5	Retained classifier	83.83 ± 0.76	75.71 ± 2.77	9/9	82.03 ± 1.38	77.69 ± 1.64	9/9
5	No classifier (recalibrated)	83.55 ± 0.29	62.19 ± 6.29	7/9	81.21 ± 1.18	73.83 ± 6.90	7/9
5	Anchored text + prototypes	83.83 ± 0.76	73.28 ± 3.14	9/9	81.95 ± 0.54	75.45 ± 1.14	8/9
5	Dynamic reference	83.83 ± 0.76	73.83 ± 0.74	9/9	82.44 ± 1.94	76.58 ± 2.18	9/9

Table S14: Oxford-IIIT Pets: final-session outcomes under the same repeated-calibration policies and aggregation as Table S13.

Shots	Readout	Match base	Match novel	Feasible	Val. base	Val. novel	Feasible
1	Retained classifier	92.58 ± 1.83	93.50 ± 0.89	9/9	88.07 ± 1.16	94.84 ± 0.89	9/9
1	No classifier (recalibrated)	92.58 ± 1.85	92.30 ± 2.36	9/9	88.56 ± 1.16	95.45 ± 0.70	6/9
1	Anchored text + prototypes	92.57 ± 1.84	93.03 ± 1.60	9/9	87.59 ± 1.34	94.59 ± 1.55	9/9
1	Dynamic reference	92.56 ± 1.82	93.74 ± 1.02	9/9	87.89 ± 1.50	95.47 ± 0.16	8/9
2	Retained classifier	92.28 ± 2.51	94.42 ± 1.14	9/9	88.05 ± 2.52	95.66 ± 0.49	9/9
2	No classifier (recalibrated)	92.27 ± 2.52	93.97 ± 0.96	9/9	87.04 ± 3.11	95.89 ± 0.27	7/9
2	Anchored text + prototypes	92.27 ± 2.50	94.33 ± 0.58	9/9	87.64 ± 2.55	95.51 ± 0.58	9/9
2	Dynamic reference	92.27 ± 2.50	94.62 ± 1.31	9/9	87.85 ± 2.40	95.77 ± 0.27	9/9
5	Retained classifier	92.40 ± 2.41	94.95 ± 1.39	9/9	88.31 ± 2.40	96.06 ± 0.47	9/9
5	No classifier (recalibrated)	92.40 ± 2.42	94.08 ± 0.92	9/9	87.39 ± 2.71	96.12 ± 0.41	7/9
5	Anchored text + prototypes	92.39 ± 2.39	94.79 ± 0.92	9/9	87.94 ± 2.68	95.99 ± 0.80	9/9
5	Dynamic reference	92.40 ± 2.40	94.84 ± 1.36	9/9	87.44 ± 2.34	96.12 ± 0.27	9/9

Table S15: ImageNet-R: final-session outcomes under the same repeated-calibration policies and aggregation as Table S13.

Shots	Readout	Match base	Match novel	Feasible	Val. base	Val. novel	Feasible
1	Retained classifier	86.24 ± 0.38	81.59 ± 0.50	9/9	85.73 ± 0.71	81.95 ± 0.67	9/9
1	No classifier (recalibrated)	86.25 ± 0.37	70.69 ± 10.24	9/9	86.13 ± 0.37	69.56 ± 13.12	9/9
1	Anchored text + prototypes	86.24 ± 0.36	78.57 ± 0.59	9/9	86.14 ± 0.54	78.97 ± 0.47	9/9
1	Dynamic reference	86.25 ± 0.36	81.78 ± 0.76	9/9	85.87 ± 0.61	82.11 ± 1.07	9/9
2	Retained classifier	86.38 ± 0.34	82.67 ± 1.21	9/9	86.10 ± 0.63	82.86 ± 1.30	9/9
2	No classifier (recalibrated)	86.38 ± 0.33	69.12 ± 14.06	9/9	86.27 ± 0.38	58.73 ± 33.47	9/9
2	Anchored text + prototypes	86.39 ± 0.34	81.53 ± 2.15	9/9	86.40 ± 0.40	81.50 ± 1.68	9/9
2	Dynamic reference	86.38 ± 0.33	82.80 ± 0.65	9/9	86.04 ± 0.67	83.02 ± 0.76	9/9
5	Retained classifier	86.35 ± 0.27	84.48 ± 0.35	9/9	86.08 ± 0.44	84.69 ± 0.46	9/9
5	No classifier (recalibrated)	86.34 ± 0.27	73.89 ± 9.72	9/9	86.36 ± 0.21	63.98 ± 26.37	9/9
5	Anchored text + prototypes	86.34 ± 0.27	83.45 ± 1.62	9/9	86.23 ± 0.52	83.65 ± 1.19	9/9
5	Dynamic reference	86.35 ± 0.29	84.42 ± 0.56	9/9	86.21 ± 0.47	84.50 ± 0.69	9/9

Appendix G.3. Readout cost and restart verification

Table S16 measures 512 cached queries on the RTX 4070 Laptop in float32. Each readout receives five warm-ups and 20 synchronized timing repetitions. Static cache values are constructed before timing, query chunks contain 256 images for retrieval, and unused score branches are skipped. An independently checked implementation verifies score agreement within 10^{-3} for every optimized readout. Persistent tensor bytes count text anchors, learned weights and references/prototypes for the tables, or text anchors, cache keys and the stored dense one-hot/class-mean value matrix for retrieval. These are implementation-level readout costs, excluding the common image encoder, query features and temporary workspaces. They do not represent complete model memory or end-to-end image inference time.

Every calibration and arrival-session result is written atomically. A live pause/resume check preserved a completed session's SHA-256 hash and modification time, then completed all 90 configuration units. The numerical verifier reconstructs group predictions, macro weights, selected thresholds and test metrics from archived sufficient statistics, including the base-validation selection arrays. The trial-zero five-shot table results agree exactly with the prior three-dataset records. Source, protocol and cache hashes, all outcomes and reproduction commands accompany the release.

Table S16: Measured readout latency and persistent tensor storage for a final-session 512-query batch on the same laptop GPU. MiB denotes 2^{20} bytes. Cache keys number 4,937 on CIFAR, 553 on Pets and 3,830 on ImageNet-R in partition zero.

Data	Readout	Median (ms)	P90 (ms)	Stored tensors (MiB)
CIFAR-100	Retained classifier	0.715	0.763	0.586
CIFAR-100	Dense text + prototypes	0.515	0.542	0.469
CIFAR-100	Tip-Adapter sum	0.838	0.862	16.205
CIFAR-100	Tip-Adapter mean	0.841	0.857	16.205
Pets	Retained classifier	0.880	1.280	0.220
Pets	Dense text + prototypes	0.586	0.680	0.176
Pets	Tip-Adapter sum	0.564	0.673	1.771
Pets	Tip-Adapter mean	0.569	0.732	1.771
ImageNet-R	Retained classifier	0.864	0.918	1.465
ImageNet-R	Dense text + prototypes	0.631	0.790	1.172
ImageNet-R	Tip-Adapter sum	0.759	0.873	14.729
ImageNet-R	Tip-Adapter mean	0.732	0.898	14.729

Appendix H. Validation-query budget for admission thresholds

This experiment varies the old-class query labels used to select the admission threshold, conditional on the original five-shot classifier and fusion calibration. All fitted parameters, cache settings and branch mixtures stay fixed. The evaluation uses the original trial-zero score archives, class partitions and arrival order zero on each dataset. No image encoder or new classifier is fitted.

For each base class with n_c validation images, the nominal fraction f keeps $\max(1, \lfloor f n_c \rfloor)$ images. Fractions $f \in \{0.25, 0.5\}$ use three predeclared subset seeds, with nested class-wise permutations; the full budget uses all images once. Actual query counts are reported because rounding and the one-query minimum change the realized fractions. Macro weights are recomputed from the selected class counts. The retained readout's closed-base accuracy on that subset defines a shared target $rA^{\text{closed}}_{B,\text{ref},\text{subset}}$ for all six readouts, with $r \in \{0.95, 0.99\}$. The largest feasible offset uses only the selected base-validation margins and labels.

After selection, test queries measure both group accuracies. A test target is met when $A_{B,\text{test}} \geq rA^{\text{closed}}_{B,\text{ref},\text{test}}$; this criterion is evaluated after the offset is fixed and never enters threshold selection. Table S17 separates validation feasibility from test target attainment. Subset repeats are averaged within each class partition before the mean and sample standard deviation over available partition means. The nine reduced-budget cases share three underlying partitions and are not independent repetitions. Small per-class counts quantize the selected threshold, so different target ratios can yield identical predictions and

increasing the validation fraction need not change test target attainment monotonically. These empirical policies do not supply a finite-sample guarantee on test retention.

Table S17: Retained readout under different validation-query budgets after the final arrival. Accuracies are percentages, mean ± partition-level sample standard deviation after averaging subset repeats. Queries gives the actual old-class validation count range. Val. feasible counts partition/subset combinations meeting the validation constraint; Test met counts feasible combinations meeting the separate test criterion. The full budget uses one subset per partition.

Data	Nominal budget	Queries	Target	Base test	Novel test	Val. feasible	Test met
CIFAR-100	25%	40	95%	82.46 ± 2.25	75.14 ± 3.50	9/9	4/9
CIFAR-100	25%	40	99%	85.21 ± 2.64	66.99 ± 9.18	9/9	4/9
CIFAR-100	50%	68	95%	81.94 ± 3.23	76.92 ± 3.72	9/9	3/9
CIFAR-100	50%	68	99%	85.85 ± 2.61	64.30 ± 10.71	9/9	5/9
CIFAR-100	100%	161–163	95%	82.99 ± 2.29	75.71 ± 3.14	3/3	1/3
CIFAR-100	100%	161–163	99%	86.48 ± 1.65	62.83 ± 10.11	3/3	2/3
Pets	25%	15	95%	91.77 ± 2.39	92.42 ± 2.64	9/9	5/9
Pets	25%	15	99%	91.77 ± 2.39	92.42 ± 2.64	9/9	4/9
Pets	50%	25	95%	92.02 ± 0.95	94.75 ± 1.23	9/9	5/9
Pets	50%	25	99%	95.05 ± 0.48	90.49 ± 5.09	9/9	5/9
Pets	100%	60	95%	90.29 ± 2.90	95.68 ± 0.83	3/3	0/3
Pets	100%	60	99%	96.43 ± 2.15	88.67 ± 7.89	3/3	2/3
ImageNet-R	25%	696–731	95%	85.85 ± 0.38	84.98 ± 0.38	9/9	2/9
ImageNet-R	25%	696–731	99%	89.79 ± 0.16	74.81 ± 2.33	9/9	5/9
ImageNet-R	50%	1444–1519	95%	85.93 ± 0.65	84.87 ± 0.63	9/9	1/9
ImageNet-R	50%	1444–1519	99%	89.90 ± 0.33	74.24 ± 2.68	9/9	5/9
ImageNet-R	100%	2938–3087	95%	86.01 ± 0.59	84.73 ± 0.71	3/3	0/3
ImageNet-R	100%	2938–3087	99%	89.94 ± 0.25	74.48 ± 1.35	3/3	2/3

The full-budget policies reproduce the original validation-selected outcomes exactly, including their feasibility. Saved test and validation predictions independently reconstruct every new selected offset, macro-weighted result and target-attainment flag. The 315 atomic session units contain 3,780 metric records across all five arrivals. Repeating the run preserves completed file hashes and modification times. The complete 50%/100% comparator outcomes, actual retained fractions and achieved test-retention ratios are provided in `validation_budget_summary.csv` and the raw records. This experiment measures threshold-selection label demand; the initial full-data fusion calibration remains fixed throughout.

Table S18: All readouts at the nominal 25% validation-query budget and 95% target after the final arrival. All classifiers, mixtures and cache settings are held fixed from the original five-shot calibration. Accuracy aggregation and feasibility/attainment counts follow Table S17. Missing metrics correspond to zero feasible subsets.

Data	Readout	Base test	Novel test	Val. feasible	Test met
CIFAR-100	Retained classifier	82.46 ± 2.25	75.14 ± 3.50	9/9	4/9
CIFAR-100	Fixed-gate removal	79.39 ± 2.08	74.84 ± 0.15	5/9	1/5
CIFAR-100	Recalibrated removal	79.48 ± 2.19	73.22 ± 2.45	5/9	1/5
CIFAR-100	Dense text + prototypes	81.30 ± 2.08	77.12 ± 2.64	7/9	0/7
CIFAR-100	Tip-Adapter sum	79.32 ± 2.30	71.40 ± 7.86	6/9	1/6
CIFAR-100	Tip-Adapter mean	79.51 ± 3.58	77.12 ± 5.00	8/9	1/8
Pets	Retained classifier	91.77 ± 2.39	92.42 ± 2.64	9/9	5/9
Pets	Fixed-gate removal	88.87 ± 3.95	92.77 ± 5.91	6/9	1/6
Pets	Recalibrated removal	87.68 ± 3.95	89.64 ± 11.35	6/9	1/6
Pets	Dense text + prototypes	88.50 ± 6.27	93.41 ± 3.82	7/9	2/7
Pets	Tip-Adapter sum	89.86 ± 3.58	89.59 ± 5.68	7/9	2/7
Pets	Tip-Adapter mean	90.25 ± 1.80	91.90 ± 3.31	7/9	2/7
ImageNet-R	Retained classifier	85.85 ± 0.38	84.98 ± 0.38	9/9	2/9
ImageNet-R	Fixed-gate removal	85.66 ± 0.86	79.23 ± 1.90	7/9	3/7
ImageNet-R	Recalibrated removal	85.74 ± 0.90	79.08 ± 1.74	7/9	3/7
ImageNet-R	Dense text + prototypes	86.39 ± 0.50	83.05 ± 0.22	9/9	4/9
ImageNet-R	Tip-Adapter sum	85.47 ± 0.89	77.71 ± 1.44	8/9	4/8
ImageNet-R	Tip-Adapter mean	86.49 ± 0.54	81.22 ± 0.63	9/9	6/9

Appendix I. Ten-partition replication, encoder transfer and simple fusion

The expanded experiment fixes class-partition seeds $0, \ldots, 9$ on each of the three existing image pools. Assignments and pseudo splits $0, 1, 2$ are retained exactly, and the original L/14 results reconstruct with maximum metric difference zero. New seeds use the same frequency-stratified allocation, base/novel counts, five-shot reserve, pseudo-episode selection and two arrival-order rules. These are ten distinct class assignments on shared images, not ten independent datasets. Primary fixed-gate comparisons average both orders within a partition. The conditional paired percentile bootstrap uses 20,000 resamples of the ten paired partition values with seed 2026. Constrained differences use identical jointly feasible partition/order/session cells. JSON records retain the partition values and available sets, and CSV tables report every budget.

Appendix I.1. Second encoder and access controls

OpenAI CLIP ViT-B/32 uses cached revision `3d74acf9a28c67741b2f4f2ea7635f0aaf6f0268` with 512-dimensional projected features; L/14 uses 768 dimensions. Both consume the exact same fitting, validation, support and test images, dataset-specific preprocessing and eight text templates. Text anchors and ridge columns are recomputed for the selected

encoder; no task-adapted features are used. The two encoders have separate closed-base references and hence separate absolute targets. Config, weights, tokenizer and pool hashes are recorded. Batch-16 feature shards support atomic restart with contiguous-index guards, and eight fresh test images are re-encoded per dataset to check each completed B/32 cache.

Appendix I.2. Raw and fixed-temperature controls

Raw fusion uses q_T, q_L and q_P directly in the same availability pattern as the retained table. Fixed-temperature fusion uses q_e/σ_e, where

$$\sigma_e = \max\left(10^{-6}, \frac{1}{|\mathcal{V}_B|}\sum_{x\in\mathcal{V}_B} \mathrm{Std}_{r\in\mathcal{R}_e}\, q_e(x,r)\right).$$

For the pseudo episode, $\mathcal{V}_B$ contains base-validation queries and $\mathcal{R}_e$ contains pseudo-base reference columns. Full-model scales use full base columns and base validation before arrivals. The three scalars remain fixed across sessions. Coefficient and offset grids are identical to the main retained model, with the same harmonic pseudo-episode objective. Neither control requires current-vocabulary statistics; both can preserve old scores. Their performance tests the empirical value of the per-query normalization, conditional on this specified calibration grid.

Appendix I.3. Full matched and validation-policy results

The main validation table includes retained, recalibrated-removal, dense-prototype, class-mean cache and simple-fusion alternatives. Summed cache and fixed-gate removal remain in the complete records. The following tables give both encoders' matched-test frontiers and the B/32 validation policy at the 95% budget. Every feasible set and test-attainment count refers to the final active vocabulary; order zero counts each partition once. Because means may cover different feasible sets, an unpaired difference of available means is not reported as a paired gain. Complete 80/85/90/95/99% outcomes are retained in `revision_summary.csv`; per-partition raw predictions supply both test and validation sufficient statistics.

Table S19: Complete eight-readout ViT-L/14 validation-selected 95% policy. Accuracies are achieved test percentages, mean ± sample standard deviation over feasible partitions. Feasible and Test met have the same meanings as the main validation-policy table.

Data	Readout	Base	Novel	Feasible	Test met
CIFAR-100	Retained references	84.26 ± 1.91	77.10 ± 2.53	10/10	7/10
CIFAR-100	Fixed-gate removal	83.05 ± 1.64	65.32 ± 12.41	6/10	1/6
CIFAR-100	Recalibrated removal	82.91 ± 1.62	66.03 ± 13.04	6/10	1/6
CIFAR-100	Dense text + prototypes	84.00 ± 1.76	75.10 ± 7.15	8/10	4/8
CIFAR-100	Raw-score fusion	82.11 ± 2.51	64.45 ± 13.35	7/10	1/7
CIFAR-100	Fixed temperatures	83.72 ± 2.24	70.93 ± 10.56	10/10	5/10
CIFAR-100	Tip-Adapter sum	83.21 ± 1.71	65.36 ± 12.68	5/10	0/5
CIFAR-100	Tip-Adapter mean	83.19 ± 1.72	68.69 ± 10.39	9/10	2/9
Pets	Retained references	89.95 ± 2.67	93.79 ± 3.83	10/10	1/10
Pets	Fixed-gate removal	90.43 ± 3.16	93.81 ± 3.11	8/10	3/8
Pets	Recalibrated removal	90.48 ± 3.04	93.44 ± 3.56	8/10	3/8
Pets	Dense text + prototypes	89.46 ± 3.08	94.47 ± 2.60	10/10	2/10
Pets	Raw-score fusion	91.35 ± 2.91	92.72 ± 2.70	9/10	4/9
Pets	Fixed temperatures	90.48 ± 3.42	94.21 ± 3.19	9/10	3/9
Pets	Tip-Adapter sum	89.18 ± 1.73	91.61 ± 6.00	7/10	1/7
Pets	Tip-Adapter mean	90.73 ± 2.17	91.39 ± 4.32	10/10	4/10
ImageNet-R	Retained references	86.15 ± 0.94	84.63 ± 0.95	10/10	2/10
ImageNet-R	Fixed-gate removal	86.17 ± 0.65	66.89 ± 22.80	10/10	3/10
ImageNet-R	Recalibrated removal	86.16 ± 0.71	66.88 ± 22.79	10/10	3/10
ImageNet-R	Dense text + prototypes	86.01 ± 0.94	83.94 ± 1.10	10/10	3/10
ImageNet-R	Raw-score fusion	86.15 ± 0.68	65.55 ± 20.42	10/10	2/10
ImageNet-R	Fixed temperatures	85.99 ± 0.71	82.92 ± 1.03	10/10	1/10
ImageNet-R	Tip-Adapter sum	86.06 ± 0.78	74.52 ± 3.69	10/10	2/10
ImageNet-R	Tip-Adapter mean	86.08 ± 1.08	81.61 ± 1.51	10/10	4/10

Table S20: ViT-L/14 matched-test 95% retention frontier across ten class partitions. Values are percentages, mean ± sample standard deviation over feasible partitions after averaging orders. Test met equals feasibility by construction in this descriptive sweep.

Data	Readout	Base	Novel	Feasible	Test met
CIFAR-100	Retained references	83.91 ± 1.42	78.02 ± 2.27	10/10	10/10
CIFAR-100	Fixed-gate removal	83.25 ± 1.17	61.68 ± 9.35	6/10	6/6
CIFAR-100	Recalibrated removal	83.25 ± 1.17	63.78 ± 6.52	6/10	6/6
CIFAR-100	Dense text + prototypes	83.91 ± 1.42	76.18 ± 2.56	10/10	10/10
CIFAR-100	Raw-score fusion	83.25 ± 1.17	62.37 ± 6.38	6/10	6/6
CIFAR-100	Fixed temperatures	83.64 ± 1.21	74.53 ± 2.70	9/10	9/9
CIFAR-100	Tip-Adapter sum	83.16 ± 1.28	61.83 ± 4.92	5/10	5/5
CIFAR-100	Tip-Adapter mean	83.91 ± 1.42	65.20 ± 11.69	10/10	10/10
Pets	Retained references	91.56 ± 1.87	93.30 ± 3.74	10/10	10/10
Pets	Fixed-gate removal	91.57 ± 1.88	93.78 ± 2.17	10/10	10/10
Pets	Recalibrated removal	91.56 ± 1.88	93.32 ± 2.10	10/10	10/10
Pets	Dense text + prototypes	91.56 ± 1.87	93.79 ± 2.78	10/10	10/10
Pets	Raw-score fusion	91.57 ± 1.88	92.39 ± 2.75	10/10	10/10
Pets	Fixed temperatures	91.56 ± 1.87	93.64 ± 2.48	10/10	10/10
Pets	Tip-Adapter sum	91.51 ± 1.98	87.78 ± 9.98	9/10	9/9
Pets	Tip-Adapter mean	91.57 ± 1.87	92.02 ± 4.15	10/10	10/10
ImageNet-R	Retained references	86.42 ± 0.53	84.31 ± 1.01	10/10	10/10
ImageNet-R	Fixed-gate removal	86.41 ± 0.56	73.82 ± 6.70	9/10	9/9
ImageNet-R	Recalibrated removal	86.42 ± 0.56	73.50 ± 6.79	9/10	9/9
ImageNet-R	Dense text + prototypes	86.42 ± 0.53	83.02 ± 1.40	10/10	10/10
ImageNet-R	Raw-score fusion	86.42 ± 0.56	70.85 ± 8.03	9/10	9/9
ImageNet-R	Fixed temperatures	86.43 ± 0.53	82.49 ± 1.11	10/10	10/10
ImageNet-R	Tip-Adapter sum	86.43 ± 0.56	73.16 ± 5.66	9/10	9/9
ImageNet-R	Tip-Adapter mean	86.43 ± 0.52	80.91 ± 1.48	10/10	10/10

Table S21: ViT-B/32 matched-test 95% frontier under the same access, aggregation and feasibility reporting as Table S20. Absolute targets are encoder-specific.

Data	Readout	Base	Novel	Feasible	Test met
CIFAR-100	Retained references	76.16 ± 1.99	59.64 ± 2.10	10/10	10/10
CIFAR-100	Fixed-gate removal	74.84 ± 1.75	49.81 ± 10.48	4/10	4/4
CIFAR-100	Recalibrated removal	74.84 ± 1.75	49.81 ± 10.48	4/10	4/4
CIFAR-100	Dense text + prototypes	76.15 ± 2.22	51.78 ± 8.42	8/10	8/8
CIFAR-100	Raw-score fusion	75.12 ± 1.63	46.30 ± 9.93	5/10	5/5
CIFAR-100	Fixed temperatures	75.71 ± 1.46	54.14 ± 4.58	9/10	9/9
CIFAR-100	Tip-Adapter sum	74.08 ± 1.05	52.81 ± 5.29	3/10	3/3
CIFAR-100	Tip-Adapter mean	76.16 ± 2.11	46.50 ± 11.77	9/10	9/9
Pets	Retained references	86.92 ± 2.17	76.54 ± 6.71	10/10	10/10
Pets	Fixed-gate removal	86.45 ± 2.27	77.62 ± 6.10	5/10	5/5
Pets	Recalibrated removal	86.46 ± 2.29	78.61 ± 5.07	5/10	5/5
Pets	Dense text + prototypes	86.84 ± 2.33	71.48 ± 15.80	8/10	8/8
Pets	Raw-score fusion	87.12 ± 2.19	67.25 ± 11.41	9/10	9/9
Pets	Fixed temperatures	86.93 ± 2.16	74.97 ± 6.20	10/10	10/10
Pets	Tip-Adapter sum	86.83 ± 2.22	70.08 ± 9.72	6/10	6/6
Pets	Tip-Adapter mean	87.28 ± 2.29	68.64 ± 21.10	8/10	8/8
ImageNet-R	Retained references	70.38 ± 0.87	56.36 ± 1.11	10/10	10/10
ImageNet-R	Fixed-gate removal	69.72	31.36	1/10	1/1
ImageNet-R	Recalibrated removal	69.71	30.54	1/10	1/1
ImageNet-R	Dense text + prototypes	70.37 ± 0.88	48.30 ± 3.45	10/10	10/10
ImageNet-R	Raw-score fusion	69.71	28.06	1/10	1/1
ImageNet-R	Fixed temperatures	70.38 ± 0.87	53.72 ± 1.35	10/10	10/10
ImageNet-R	Tip-Adapter sum	69.67 ± 0.03	35.63 ± 8.58	2/10	2/2
ImageNet-R	Tip-Adapter mean	70.45 ± 0.92	41.88 ± 7.58	8/10	8/8

Table S22: ViT-B/32 validation-selected 95% policy. Values are achieved base/novel test percentages over validation-feasible partitions; Test met counts those meeting the independent test-retention criterion. Validation feasibility does not guarantee test target attainment.

Data	Readout	Base	Novel	Feasible	Test met
CIFAR-100	Retained references	75.98 ± 2.43	59.04 ± 3.66	10/10	5/10
CIFAR-100	Fixed-gate removal	73.46 ± 1.73	52.54 ± 10.71	7/10	1/7
CIFAR-100	Recalibrated removal	73.40 ± 1.79	54.20 ± 6.94	7/10	1/7
CIFAR-100	Dense text + prototypes	74.30 ± 2.85	55.73 ± 8.03	7/10	2/7
CIFAR-100	Raw-score fusion	73.82 ± 2.67	49.47 ± 12.37	8/10	2/8
CIFAR-100	Fixed temperatures	73.90 ± 2.39	56.16 ± 8.19	9/10	1/9
CIFAR-100	Tip-Adapter sum	73.55 ± 2.53	49.25 ± 12.78	7/10	1/7
CIFAR-100	Tip-Adapter mean	73.54 ± 2.88	57.47 ± 8.46	7/10	0/7
Pets	Retained references	88.16 ± 3.34	72.75 ± 8.65	10/10	8/10
Pets	Fixed-gate removal	86.17 ± 2.74	76.30 ± 6.46	5/10	3/5
Pets	Recalibrated removal	86.24 ± 2.14	77.70 ± 5.37	5/10	2/5
Pets	Dense text + prototypes	86.89 ± 3.09	70.27 ± 15.78	7/10	3/7
Pets	Raw-score fusion	86.93 ± 5.36	63.79 ± 15.01	9/10	5/9
Pets	Fixed temperatures	85.53 ± 5.27	74.57 ± 7.40	10/10	4/10
Pets	Tip-Adapter sum	86.87 ± 2.42	73.74 ± 4.16	4/10	2/4
Pets	Tip-Adapter mean	84.97 ± 3.13	81.94 ± 3.51	5/10	1/5
ImageNet-R	Retained references	70.18 ± 0.96	56.80 ± 1.33	10/10	3/10
ImageNet-R	Fixed-gate removal	68.23 ± 0.84	19.12 ± 6.82	2/10	0/2
ImageNet-R	Recalibrated removal	68.23 ± 0.84	19.12 ± 6.82	2/10	0/2
ImageNet-R	Dense text + prototypes	70.43 ± 1.50	46.95 ± 4.97	10/10	6/10
ImageNet-R	Raw-score fusion	68.16 ± 0.84	21.79 ± 6.87	2/10	0/2
ImageNet-R	Fixed temperatures	70.46 ± 1.26	53.38 ± 1.43	10/10	6/10
ImageNet-R	Tip-Adapter sum	68.89 ± 0.79	38.12 ± 2.27	3/10	0/3
ImageNet-R	Tip-Adapter mean	69.99 ± 1.61	46.39 ± 3.62	9/10	4/9

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